

Complex scaling for open waveguides

Charles L. Epstein
Center for Computational Mathematics
Flatiron Institute
New York, NY 10010
`cepstein@flatironinstitute.org`

Tristan Goodwill
Department of Statistics and CCAM
University of Chicago
Chicago, IL 60637
`tgoodwill@uchicago.edu`

Jeremy Hoskins
Department of Statistics and CCAM
University of Chicago
Chicago, IL 60637
`jeremyhoskins@uchicago.edu`

Solomon Quinn
Center for Computational Mathematics
Flatiron Institute
New York, NY 10010
`squinn@flatironinstitute.org`

Manas Rachh
Center for Computational Mathematics
Flatiron Institute
New York, NY 10010
`mrachh@flatironinstitute.org`

Abstract

In this work we analyze the complex scaling method applied to the problem of time-harmonic scalar wave propagation in junctions between ‘leaky,’ or open dielectric waveguides. In [20, 21, 23, 24], it was shown that under suitable assumptions the problem can be reduced to a system of Fredholm second-kind integral equations on an infinite interface, transverse to the waveguides. Here, we show that the kernels appearing in the integral equation admit a rapidly decaying analytic continuation on certain natural totally real submanifolds of $\mathbb{C}^2$. We then show that for suitable, physically-meaningful, boundary data the resulting solutions to the integral equations themselves admit analytic continuation and satisfy related asymptotic estimates. By deforming the integral equation to a suitable contour, the decay in the kernels, density, and data enable straightforward discretization and truncation, with an error that decays exponentially in the truncation length. We illustrate our results with several representative numerical examples.

Keywords:

Open waveguides, scattering, Fredholm integral equations, complexification, infinite boundary, outgoing solution

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1 Introduction

Waveguides are important and ubiquitous tools for modeling acoustic, electromagnetic, and fluidic problems. Their role in long distance transmission of signals makes them ideal for use in a wide variety of applications, *inter alia*, photonic circuits [7, 30, 31, 34, 36, 38, 44, 50, 54], medical devices [11, 25, 39, 47, 52, 53, 57], fiber optics [1, 19, 29, 41, 45, 46], and loudspeakers [6, 28, 32, 37, 55]. In many contexts the waveguides are modeled as closed systems — not coupled in any way to the surrounding material. Still, in a number of situations, particularly in photonic devices, coupling

with the surrounding medium is non-trivial and an important source of signal degradation. This coupling poses significant challenges both analytically and numerically. Essentially, the radiated field in the surrounding medium decays slowly in space, which, when combined with the non-compact nature of the waveguides, renders approaches based on truncation difficult or impossible.

In two dimensions the basic mathematical model is the following partial differential equation (PDE)

$$[\Delta + k^2(1 + \chi_{(-\infty,0)}(x_1)q_l(x_2) + \chi_{[0,\infty)}(x_1)q_r(x_2))]u(\mathbf{x}) = f(\mathbf{x}), \quad (1.1)$$

where $\mathbf{x} = (x_1, x_2)$, $k \in \mathbb{R}$ denotes the wavenumber of the ambient material, and the functions q_l, q_r are compactly supported piecewise smooth functions that encode the material properties of the left and right waveguide respectively. For uniqueness, the above PDE must be supplemented with suitable conditions at infinity, with corresponding constraints placed on the source term f . Though fairly intuitive, the full description of these radiation conditions is rather involved, and we refer the interested reader to [23].

In [20, 21, 23] the equation (1.1) is analyzed by reducing it to a system of boundary integral equations (BIEs). More precisely, the plane is divided into two halves, with their interface chosen so as to cut the waveguides orthogonally along their intersection, i.e., the line $\{x_1 = 0\}$. By introducing suitable layer potentials generated by densities on the interface, it was shown that the resulting boundary integral equation (BIE) for these densities has unique solutions in suitable function spaces. In principle one can use this formulation directly to solve (1.1). In practice the decay of the densities on this interface, and the decay of the kernels of the integral operators, is too slow to make such an approach practical. In the subsequent paper [24], building on the method of [35], it was shown empirically that by introducing *complex scaling* along the interface, i.e., replacing the line $\{x_1 = 0\}$ by any curve in a family of totally real submanifolds of $\{0\} \times \mathbb{C} \subset \mathbb{R} \times \mathbb{C}$, this boundary integral equation can be solved accurately and efficiently on a bounded domain. Here, extending the method of [22], and the analytic approach of [22], we give a rigorous proof of the validity of this approach. In particular, we show that on suitably chosen deformed contours, the analytic continuation of the BIEs of [20, 21, 23] admit unique, and rapidly decaying solutions. Additionally, when these contours are truncated, as is necessary for numerical computations, the resulting approximations to the solutions of (1.1) converge rapidly to the true solution. We further extend the method of [24] to account for waveguides with material parameters that are piecewise smooth in the transverse direction.

One proposed approach for solving problems of this class is via domain decomposition, discretizing the problem partially in the Fourier domain [16]. For simple problems one can compute an operator which takes ‘right going’ to ‘left going’ waves for each half-space. The solution to the full problem can then be obtained by enforcing continuity across the interface. Though interesting analytically, these methods have yet to see practical numerical realization, and are relatively inflexible in the presence of perturbations and non-perpendicular waveguides. An alternate philosophy is instead to discretize only the boundary of the waveguide. Though the analytic structure of the equation (1.1) prevents straightforward truncation of the resulting BIEs, several methods have been developed to produce accurate solutions on suitably truncated problems. One approach, referred to as *windowed Green’s functions*, uses partitions of unity to discretize the waveguides near the boundary in physical space, and the infinite flat part of the geometry in Fourier space [9, 33]. Though these methods are empirically effective in an array of contexts, they do not directly address the solvability of the integral equation on the untruncated domain, nor its truncated counterpart.

A related set of methods involve introduction of perfectly matched layers (PMLs), which frequently arise in the restriction of PDEs defined on infinite domains to compact regions. These methods use the outgoing nature of the solutions to extend them to a two dimensional totally real submanifold of $\mathbb{C}^2$ where the solution is exponentially decaying and can be effectively truncated [5, 12, 18]. Recently there has also been work on integral equation based PML methods [35], and our method is of this general character. As yet, it is still an open problem to apply these techniques to BIE formulations of non-compact waveguide problems such as (1.1).

In this paper, as in [24], we consider a different approach, solving on the fictitious interface as in [20, 21, 23] rather than on the boundary of the waveguide itself. By ‘complexifying’ the coordinates of this artificial interface, we prove that one obtains an invertible Fredholm integral equation of second kind. The solutions of this new integral equation can then be used to recover the solution of

the original equation (1.1). Significantly, the solutions of this ‘complexified’ BIE decay exponentially, enabling straightforward truncation to be applied in numerical implementations. For a given choice of complexification, which depends on the choice of a positive number L_C , the solution of the BIE can then be used to calculate the solution $u(\mathbf{x})$ for $\{\mathbf{x} : |x_2| \leq L_C\}$.

The remainder of the paper is as follows. In Section 2, we review the construction of the BIEs in [20, 21, 23]. Following this, in Section 3 we state our main analytic result. Sections 4, and 5 contain the proofs of these main results. Finally, in Section 7 we briefly describe a numerical method for approximately solving the complexified BIEs, extending the work in [24], to allow for piecewise smooth (in the transverse coordinate) waveguide parameters.

2 Boundary integral equation for interface matching

Following [20, 21, 23], we solve the PDE (1.1) by first replacing it with a transmission problem, which is then reduced to a system of boundary integral equations on a fictitious interface separating the two semi-infinite waveguides. We assume that $q_{l,r}$ are piecewise continuous and supported on the interval $[-d, d]$. For $\Im k > 0$ we let $G_k^{l,r}$ denote the kernels of the inverses of the self adjoint partial differential operators

$$\Delta_x G_k^{l,r}(\mathbf{x}; \mathbf{y}) + k^2(1 + q_{l,r}(x_2))G_k^{l,r}(\mathbf{x}; \mathbf{y}) = \delta(\mathbf{x} - \mathbf{y}). \quad (2.1)$$

Self adjointness implies that for k in the closed upper half plane the fundamental solution satisfies the relation

$$G_k^{l,r}(\mathbf{x}; \mathbf{y}) = \overline{G_{-\bar{k}}^{l,r}(\mathbf{y}; \mathbf{x})}.$$

Because the potential is real, these kernels are real for $k = it$, which implies that they also satisfy

$$G_k^{l,r}(\mathbf{x}; \mathbf{y}) = G_k^{l,r}(\mathbf{y}; \mathbf{x}).$$

Henceforth, we let k be a fixed positive number and set

$$G^{l,r}(\mathbf{x}; \mathbf{y}) = \lim_{\epsilon \rightarrow 0^+} G_{k+i\epsilon}^{l,r}(\mathbf{x}; \mathbf{y}). \quad (2.2)$$

That is, $G^{l,r}$ is the kernel of limiting absorption resolvent

$$\lim_{\epsilon \rightarrow 0^+} [\Delta_x + (k + i\epsilon)^2(1 + q_{l,r}(x_2))]^{-1}. \quad (2.3)$$

By continuity this kernel also satisfies the relation $G^{l,r}(\mathbf{x}; \mathbf{y}) = G^{l,r}(\mathbf{y}; \mathbf{x})$.

Using these Green’s functions we can remove the volumetric source f , reducing the problem to an auxiliary transmission problem on the line $\{(0, x_2) | x_2 \in \mathbb{R}\}$. In particular, if $\Omega_{l,r}$ denotes the left and right half-spaces then, setting

$$u_i^{l,r}(\mathbf{x}) = \int_{\mathbb{R}^2} G^{l,r}(\mathbf{x}, \mathbf{y}) f(\mathbf{y}) d\mathbf{y}, \quad \text{for } \mathbf{x} \in \Omega_{l,r}, \quad (2.4)$$

and $u^{l,r} = u - u_i^{l,r}$ on $\Omega_{l,r}$, then the functions $u^{l,r}$ satisfy

$$\begin{aligned} \Delta_x u^{l,r}(\mathbf{x}) + k^2(1 + q_{l,r}(x_2))u^{l,r}(\mathbf{x}) &= 0, \quad \text{on } \Omega_{l,r}, \\ \lim_{\epsilon \rightarrow 0^+} [u^r(\epsilon, x_2) - u^l(-\epsilon, x_2)] &= g(x_2), \quad x_2 \in \mathbb{R}, \\ \lim_{\epsilon \rightarrow 0^+} \left[\frac{\partial u^r}{\partial x_1} \Big|_{x_1=\epsilon}(x_1, x_2) - \frac{\partial u^l}{\partial x_1} \Big|_{x_1=-\epsilon}(x_1, x_2) \right] &= h(x_2), \quad x_2 \in \mathbb{R}, \end{aligned} \quad (2.5)$$

where

$$g(x_2) := - \lim_{\epsilon \rightarrow 0^+} [u_i^r(\epsilon, x_2) - u_i^l(-\epsilon, x_2)] \quad (2.6)$$

$$h(x_2) := - \lim_{\epsilon \rightarrow 0^+} \left[\frac{\partial u_i^r}{\partial x_1}(\epsilon, x_2) - \frac{\partial u_i^l}{\partial x_1}(-\epsilon, x_2) \right]. \quad (2.7)$$

In the absence of either waveguide, the fundamental solution is given by

$$G_0(\mathbf{x}; \mathbf{y}) = \frac{i}{4} H_0(k\|\mathbf{x} - \mathbf{y}\|) \in H_{\text{loc}}^{1-\epsilon}(\mathbb{R}^2),$$

for any $\epsilon > 0$, where H_0 is the zeroth order Hankel function. Defining

$$w^{l,r}(\mathbf{x}; \mathbf{y}) = G^{l,r}(\mathbf{x}; \mathbf{y}) - G_0(\mathbf{x}; \mathbf{y}), \quad (2.8)$$

it follows immediately from standard elliptic regularity theory that, for fixed $\mathbf{y}$, $w^{l,r}(\cdot; \mathbf{y}) \in H_{\text{loc}}^{3-\epsilon}(\mathbb{R}^2)$.

We next introduce the following ansatz for the left and right solutions:

$$u^{l,r}(\mathbf{x}) = \int_{-\infty}^{\infty} G^{l,r}(\mathbf{x}; 0, y_2) \tau(y_2) dy_2 - \int_{-\infty}^{\infty} \frac{\partial}{\partial y_1} \Big|_{y_1=0} G^{l,r}(\mathbf{x}; y_1, y_2) \sigma(y_2) dy_2, \quad (2.9)$$

where τ and σ are two unknown densities. Restricting this representation to $\{x_1 = 0\}$, and applying the transmission boundary conditions in (2.5), leads to the integral equations for $[\sigma, \tau]$,

$$\sigma(x_2) - \int_{-\infty}^{\infty} \frac{\partial}{\partial y_1} [w^r(0, x_2; 0, y_2) - w^l(0, x_2; 0, y_2)] \sigma(y_2) dy_2 + \quad (2.10)$$

$$\int_{-\infty}^{\infty} [w^r(0, x_2; 0, y_2) - w^l(0, x_2; 0, y_2)] \tau(y_2) dy_2 = g(x_2) \quad (2.11)$$

$$\tau(x_2) + \int_{-\infty}^{\infty} \frac{\partial}{\partial x_1} [w^r(0, x_2; 0, y_2) - w^l(0, x_2; 0, y_2)] \tau(y_2) dy_2 - \quad (2.12)$$

$$\int_{-\infty}^{\infty} \frac{\partial^2}{\partial x_1 \partial y_1} [w^r(0, x_2; 0, y_2) - w^l(0, x_2; 0, y_2)] \sigma(y_2) dy_2 = h(x_2). \quad (2.13)$$

Here we have used the decomposition of $G^{l,r}$ into $w^{l,r}$ and G_0 , together with the standard jump relations of G_0 [14]. We can make a further simplification by observing that $w^{l,r}(x_1, x_2; 0, y_2)$ and $w^{l,r}(0, x_2; y_1, y_2)$ are even functions of x_1 and y_1 , respectively. In particular, the terms involving the first derivatives of $w^{l,r}$ in (2.10) vanish.

For ease of notation, we define the operators C and D by

$$Df(x_2) := \int_{-\infty}^{\infty} k_D(x_2, y_2) f(y_2) dy_2 = \int_{-\infty}^{\infty} [w^r(0, x_2; 0, y_2) - w^l(0, x_2; 0, y_2)] f(y_2) dy_2 \quad (2.14)$$

$$Cf(x_2) = - \int_{-\infty}^{\infty} k_C(x_2, y_2) f(y_2) dy_2 = - \int_{-\infty}^{\infty} \frac{\partial^2}{\partial x_1 \partial y_1} [w^r(0, x_2; 0, y_2) - w^l(0, x_2; 0, y_2)] f(y_2) dy_2. \quad (2.15)$$

Using these definitions, we arrive at the following system of integral equations for the densities σ and τ ,

$$\begin{bmatrix} \mathcal{I} & D \\ C & \mathcal{I} \end{bmatrix} \begin{bmatrix} \sigma \\ \tau \end{bmatrix} (x_2) = \begin{bmatrix} g \\ h \end{bmatrix} (x_2), \quad \text{for all } x_2 \in \mathbb{R}. \quad (2.16)$$

Here $\mathcal{I}$ denotes the identity operator.

3 Main results

In this section we present the central analytical results of this paper. Before stating the main theorems, we first introduce some necessary notation and results related to layer potentials on complex contours. We begin by defining a certain subset of the complex plane, in which the complexification of the integral equation (2.16) is defined. Following that, we define complex extensions of the integral operators D and C from (2.14) and (2.15), respectively, together with suitable function spaces on which they act. We conclude the section with Theorem 1, which gives existence and uniqueness of

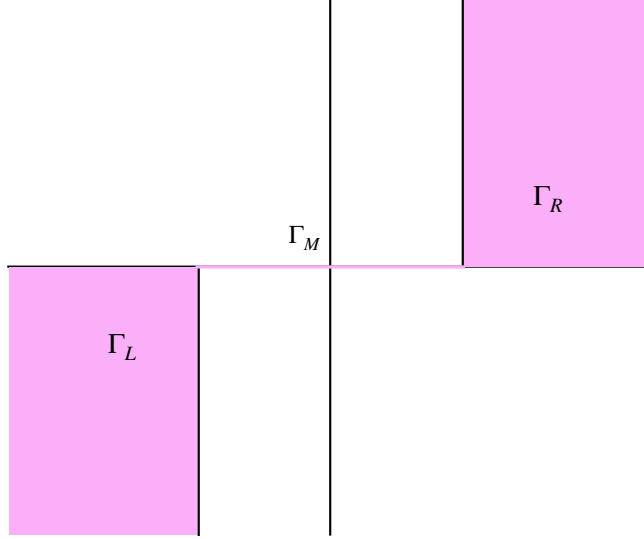


Figure 1: Plot showing Γ_C in pink.

the solution to the complexification of (2.16), Theorem 2, which connects the solutions of this complexified integral equation to solutions of (1.1) and Theorem 3, which bounds the error in computing the solutions of the complexified integral equation on truncated contours.

In our method and analysis we consider the following subsets of $\mathbb{C}$, which are convenient for defining suitable function spaces of ‘outgoing’ data.

Definition 1. Fix $L_C > d$. Define the sets $\Gamma_L, \Gamma_M, \Gamma_R \subset \mathbb{C}$, given by

$$\begin{aligned} \Gamma_L &:= \{z : \Re z < -L_C, \Im z \leq 0\}, \\ \Gamma_M &:= \{z : |\Re z| \leq L_C, \Im z = 0\}, \quad \text{and} \\ \Gamma_R &:= \{z : \Re z > L_C, \Im z \geq 0\}. \end{aligned}$$

We define Γ_C by

$$\Gamma_C := \Gamma_L \cup \Gamma_M \cup \Gamma_R,$$

see Figure 1 for an illustration.

Remark 1. Below we show that the kernel functions $k_C(x_2, y_2)$, $k_D(x_2, y_2)$ have analytic continuations to $\Gamma_C \times \Gamma_C$.

Ultimately, we wish to formulate complex extensions of (2.16) to admissible contours lying in Γ_C . The following definition provides sufficient conditions for a contour in Γ_C to be compatible with our method.

Definition 2. Let $\mathcal{G}$ denote the set of all smooth curves $\tilde{\Gamma}$ in Γ_C that satisfy the following two conditions.

(i) The curve $\tilde{\Gamma}$ is monotonic in the sense that if $x_2, y_2 \in \tilde{\Gamma}$, then

$$\Im(x_2 - y_2) \geq 0 \quad \text{implies} \quad \Re(x_2 - y_2) \geq 0. \quad (3.1)$$

(ii) For all $t \in \mathbb{R}$ there is an $x_2 \in \tilde{\Gamma}$ with $\Re(x_2) = t$.

We typically consider curves given by a parameterization $x_2(t)$ with $x_2(t) = t + i\psi(t)$, where ψ is a smooth monotonically increasing function, which is zero for $t \in [-L_C, L_C]$. We now define extensions of our integral operators over curves in $\mathcal{G}$. In the following definitions we assume that $(\sigma(y_2), \tau(y_2))$ are defined for $y_2 \in \tilde{\Gamma}$, and have sufficient decay for the integrals to be defined. Precise hypotheses are given below.

Definition 3 (Complex difference potentials). Let $\tilde{\Gamma} \in \mathcal{G}$. For every $x_2 \in \Gamma_{\mathbb{C}}$, the complexified operators on $\tilde{\Gamma}$ are given by the complex contour integrals

$$C_{\tilde{\Gamma}}[\sigma](x_2) = \int_{\tilde{\Gamma}} k_C(x_2, y_2) \sigma(y_2) dy_2 \quad (3.2)$$

and

$$D_{\tilde{\Gamma}}[\tau](x_2) = \int_{\tilde{\Gamma}} k_D(x_2, y_2) \tau(y_2) dy_2 \quad (3.3)$$

where $k_{C,D}$ are the analytic continuations of the kernels defined in (2.15) and (2.14), respectively. In the following it will also be convenient to define the system matrix $\mathcal{K}_{\tilde{\Gamma}}$ by

$$\mathcal{K}_{\tilde{\Gamma}} = \begin{bmatrix} 0 & D_{\tilde{\Gamma}} \\ C_{\tilde{\Gamma}} & 0 \end{bmatrix}. \quad (3.4)$$

We analyze the behavior of these operators on the following family of Banach spaces of piecewise analytic functions.

Definition 4. For any real numbers α and β , we define the norm

$$\|f\|_{\alpha,\beta} := \sup_{x \in \Gamma_{\mathbb{C}}} e^{\beta|\Im x|} (1 + |x|)^{\alpha} |f(x)|. \quad (3.5)$$

The space $\mathcal{C}^{\alpha,\beta}$ is the set of all continuous functions f on $\Gamma_{\mathbb{C}}$, which are analytic in $\text{int}\Gamma_L \cup \text{int}\Gamma_R$ and satisfy $\|f\|_{\alpha,\beta} < \infty$.

Remark 2. Throughout the paper, unless stated otherwise, we assume that $0 < \alpha < \frac{1}{2}$ and $\beta > -k$.

It is useful to define analogous spaces for functions defined on individual curves $\tilde{\Gamma} \in \mathcal{G}$.

Definition 5. If $\tilde{\Gamma} \in \mathcal{G}$, then for any real numbers α , and β , we define the norm

$$\|f\|_{\alpha,\beta,\tilde{\Gamma}} := \sup_{x \in \tilde{\Gamma}} e^{\beta|\Im x|} (1 + |x|)^{\alpha} |f(x)|. \quad (3.6)$$

The space $\mathcal{C}_{\tilde{\Gamma}}^{\alpha,\beta}$ is the set of all continuous functions f on $\tilde{\Gamma}$, which satisfy $\|f\|_{\alpha,\beta,\tilde{\Gamma}} < \infty$.

The fact that the $\mathcal{C}^{\alpha,\beta}$ and $\mathcal{C}_{\tilde{\Gamma}}^{\alpha,\beta}$ are Banach spaces is essentially proved in [22]. Note that

$$\mathcal{C}^{\alpha,\beta} \upharpoonright_{\tilde{\Gamma}} \subset \mathcal{C}_{\tilde{\Gamma}}^{\alpha,\beta}, \quad (3.7)$$

and the spaces $\mathcal{C}_{\mathbb{R}}^{\alpha,\beta}$ coincide with the spaces $\mathcal{C}^{\alpha}(\mathbb{R})$, which are defined in [20].

In Section 4, we prove that if the densities lie in $\mathcal{C}^{\alpha,\beta}$ then the value of the layer potential is independent of the path $\tilde{\Gamma}$ used in their definition. In Section 5, we then use this path independence to extend the existence and uniqueness theorems in [21, 23] to prove that the analytic continuation of (2.16) is well-posed. This result is summarized in the following theorem.

Theorem 1. Let $0 < \alpha < \frac{1}{2}$, and $-k < \beta \leq k$. If $[f_D, f_N] \in \mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$, then there exists a unique pair of functions $[\sigma, \tau] \in \mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$, such that for any $\tilde{\Gamma} \in \mathcal{G}$,

$$\left(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}} \right) \begin{bmatrix} \sigma \\ \tau \end{bmatrix} (x_2) = \begin{bmatrix} f_D \\ f_N \end{bmatrix} (x_2) \quad \forall x_2 \in \tilde{\Gamma}. \quad (3.8)$$

Indeed, this equation actually holds for every $x_2 \in \Gamma_{\mathbb{C}}$.

When solving equation (3.8) numerically, the densities σ, τ are obtained by solving the integral equation for $x_2 \in \tilde{\Gamma}$ for some fixed contour $\tilde{\Gamma}$. Given these densities defined on $\tilde{\Gamma}$, using the analyticity of f_D, f_N , and the analytic properties of the complex difference potentials discussed in Definition 3, we see that σ, τ extend analytically to $\Gamma_{\mathbb{C}}$ and for $x_2 \in \Gamma_{\mathbb{C}}$ can be computed via the formula:

$$\begin{bmatrix} \sigma \\ \tau \end{bmatrix} (x_2) = \begin{bmatrix} f_D \\ f_N \end{bmatrix} (x_2) - \mathcal{K}_{\tilde{\Gamma}} \begin{bmatrix} \sigma \\ \tau \end{bmatrix} (x_2). \quad (3.9)$$

The integral operator in the last term in the equation above only depends on the values of σ and τ on $\tilde{\Gamma}$.

Surprisingly, data that is incoming with specified exponential growth at ∞ in $\Gamma_{\mathbb{C}}$ is admissible for the solution of equation (3.8). This allowed growth is related to the maximum frequency of oscillations of ‘incoming’ fields in the Isozaki sense [23, 27], and ‘incoming’ waves oscillating with frequency greater than $-k$ are permissible. It should be noted that while the integral equation is uniquely solvable for such incoming data, additional restrictions may be necessary to show that the corresponding solutions to the PDE are outgoing.

In Section 5, we show that the solution of (3.8) can be used to construct a solution of the transmission problem (2.16) and thus leads to a solution of the original PDE (1.1). Before doing so, we define analytic continuations of the layer potential operators used to define $u^{l,r}$. The decay properties of the analytic continuation of $G^{l,r}$ for evaluation on $x_1 \neq 0$ are quite different from those of the kernels k_C and k_D on $x_1 = 0$. This introduces the further restriction of $\beta > 0$. In conjunction with the restriction $|x_2| < L_C$, the analytic continuation of $G^{l,r}(\mathbf{x}; 0, y_2)$ decays algebraically as a function of $|y_2|$ with a uniform rate of algebraic decay in x_1 . This decay combined with the decay of the density implies that the layer potentials on $\tilde{\Gamma}$ are absolutely convergent and thus allows us to define the analytic extensions of the layer potentials.

Definition 6 (Complex layer potentials). *Let $\tilde{\Gamma} \in \mathcal{G}$. For every $\mathbf{x} \in \mathbb{R}^2 \cap \{|x_2| < L_C\}$, the complexified operators with densities supported on $\tilde{\Gamma}$, are given by the complex contour integrals*

$$\mathcal{S}_{\tilde{\Gamma}}^{l,r}[\tau](\mathbf{x}) = \int_{\tilde{\Gamma}} G^{l,r}(\mathbf{x}; 0, y_2) \tau(y_2) dy_2 \quad (3.10)$$

and

$$\mathcal{D}_{\tilde{\Gamma}}^{l,r}[\sigma](\mathbf{x}) = \int_{\tilde{\Gamma}} \partial_{y_1} G^{l,r}(\mathbf{x}; 0, y_2) \sigma(y_2) dy_2 \quad (3.11)$$

where $G^{l,r}$ are the analytic continuations of the Green’s functions of the left and right waveguide operators.

It turns out that physically relevant scattering data arising from volumetric or point sources, and, after appropriate modifications, waveguide modes (see (6.11)) result in boundary data in $\mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$ with $\beta > 0$. Furthermore, for data arising from the waveguide modes, the contours $\tilde{\Gamma}$ must also satisfy the following slope condition in $\Gamma_{\mathbb{C}}$. $\tilde{\Gamma} \in \mathcal{G}$ is admissible if there exist $c, C > 0$ and R_0 such that for all $y_2 \in \tilde{\Gamma}$ with $|y_2| > R_0 > L_C$, we have

$$c|\Re y_2| \leq |\Im y_2| \leq C|\Re y_2|. \quad (3.12)$$

From a numerical efficiency perspective, the contours $\tilde{\Gamma}$ are anyway chosen to satisfy such criterion, and thus these conditions are effectively redundant.

The following result establishes the connection between the solution of the complexified integral equations and the solutions of the original PDE for certain choices of data.

Theorem 2. *Suppose $\beta \geq 0$ and the pair of functions $[f_D, f_N] \in \mathcal{C}^{\frac{1}{2},\beta} \oplus \mathcal{C}^{\frac{3}{2},\beta}$ satisfies*

$$\begin{aligned} f_D(x_2) &\sim \frac{e^{ik|x_2|}}{|x_2|^{\frac{1}{2}}} \sum_{l=0}^{\infty} \frac{a_l^{\pm}}{|x_2|^l} \quad \text{and} \\ f_N(x_2) &\sim \frac{e^{ik|x_2|}}{|x_2|^{\frac{3}{2}}} \sum_{l=0}^{\infty} \frac{b_l^{\pm}}{|x_2|^l}, \end{aligned} \quad (3.13)$$

as $x_2 \rightarrow \pm\infty$ with $x_2 \in \mathbb{R}$. If the pair $[\sigma_{\tilde{\Gamma}}, \tau_{\tilde{\Gamma}}] \in \mathcal{C}_{\tilde{\Gamma}}^{\alpha, \beta} \oplus \mathcal{C}_{\tilde{\Gamma}}^{\alpha + \frac{1}{2}, \beta}$ solves equation (3.8) on some $\tilde{\Gamma} \in \mathcal{G}$, then this pair has an analytic continuation to an element of $[\sigma, \tau] \in \mathcal{C}^{\frac{1}{2}, \beta} \oplus \mathcal{C}^{\frac{3}{2}, \beta}$.

The pair of functions $u^{l,r} = \mathcal{S}_{\mathbb{R}}^{l,r}[\tau] + \mathcal{D}_{\mathbb{R}}^{l,r}[\sigma]$, solves the transmission problem in (2.5), and satisfies the outgoing radiation conditions given in [23]. Moreover, for any curve $\tilde{\Gamma} \in \mathcal{G}$ satisfying the slope condition in equation (3.12), the solution constructed using the representation $u^{l,r}(\mathbf{x}) = \mathcal{S}_{\tilde{\Gamma}}^{l,r}[\tau](\mathbf{x}) + \mathcal{D}_{\tilde{\Gamma}}^{l,r}[\sigma](\mathbf{x})$ agrees with this solution for $\mathbf{x} \in \mathbb{R}^2 \cap \{|x_2| \leq L_C\}$.

Practical numerical implementation requires truncation. In the following theorem we bound the error in the computed solution, when the contour used to find the density and compute the solution is truncated.

Theorem 3. Let $0 < \alpha < \frac{1}{2}$, $0 < \beta \leq k$ and $\tilde{\Gamma}$ be a contour in $\mathcal{G}$ satisfying the slope condition in equation (3.12). Let $\tilde{\Gamma}_\epsilon$ be its truncation given by $\tilde{\Gamma}_\epsilon = \tilde{\Gamma} \cap \{x_2 \in \Gamma_{\mathbb{C}} : e^{-\beta|\Im x_2|} \geq \epsilon\}$. Suppose that f_D and f_N satisfy the conditions of the previous theorem. For sufficiently small $\epsilon > 0$ there is a unique pair $[\sigma_\epsilon, \tau_\epsilon] \in \mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}$ that solves

$$\left(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}_\epsilon}\right) \begin{bmatrix} \sigma_\epsilon \\ \tau_\epsilon \end{bmatrix} (x_2) = \begin{bmatrix} f_D \\ f_N \end{bmatrix} (x_2) \quad \forall x_2 \in \tilde{\Gamma}_\epsilon. \quad (3.14)$$

Further, there exists a $C > 0$, independent of ϵ and the data, such that

$$\left| \left(\mathcal{S}_{\tilde{\Gamma}_\epsilon}^{l,r}[\tau_\epsilon](\mathbf{x}) - \mathcal{D}_{\tilde{\Gamma}_\epsilon}^{l,r}[\sigma_\epsilon](\mathbf{x}) \right) - u^{l,r}(\mathbf{x}) \right| < C\epsilon \left(\|f_D\|_{\mathcal{C}^{\alpha, \beta}} + \|f_N\|_{\mathcal{C}^{\alpha + \frac{1}{2}, \beta}} \right) \quad (3.15)$$

for all $\mathbf{x} \in \mathbb{R}^2 \cap \{|x_2| \leq L_C\}$, where $u^{l,r}$ is the solution of (2.5) which satisfies the outgoing radiation conditions.

Remark 3. The restrictions that $\beta > 0$, and $\tilde{\Gamma}$ satisfies the slope condition in Theorems 2 and 3 can be relaxed if one restricts x_1 to also lie in a fixed interval $[-L_0, L_0]$, for any $L_0 \in (0, \infty)$. In this case we only need to require that $\beta > -k$ in order for the solution constructed via the layer potentials to agree with the solution of the transmission problem in the bounded region $(x_1, x_2) \in [-L_0, L_0] \times [-L_C, L_C]$. This also leads to estimates like those in (3.15), provided $|x_1| < L_0$.

The analytic continuation of a waveguide mode satisfies an estimate of the form $|v(x_2)| \leq Ce^{-c|\Re x_2|}$, for c, C positive constants. Because the exponential decay is in the $\Re x_2$ it is not in $\mathcal{C}^{\alpha, 0}$ space for any $\alpha > 0$. Nonetheless restricted to a curve, $\tilde{\Gamma}$, satisfying the slope condition this data does belong to $\mathcal{C}_{\tilde{\Gamma}}^{\alpha, \beta}$ for some positive α and β .

Corollary 1. Suppose $u(x_1, x_2) = v(x_2)e^{\pm i\xi x_1}$ is a waveguide mode for either q^l or q^r and the boundary data $[f_D, f_N] = [v, \pm i\xi v]$. If $\tilde{\Gamma} \in \mathcal{G}$ satisfies the slope conditions in (3.12), then there is a unique $[\tilde{\sigma}, \tilde{\tau}] \in \mathcal{C}^{1/2, k} \oplus \mathcal{C}^{1, k}$ such that

$$\left(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}}\right) \begin{bmatrix} \tilde{\sigma} + f_D \\ \tilde{\tau} + f_N \end{bmatrix} (x_2) = \begin{bmatrix} f_D \\ f_N \end{bmatrix} (x_2) \quad \forall x_2 \in \tilde{\Gamma}. \quad (3.16)$$

Further, the pair of functions

$$u^{l,r}(\mathbf{x}) = \mathcal{S}_{\mathbb{R}}^{l,r}[\tilde{\tau} + f_N](\mathbf{x}) + \mathcal{D}_{\mathbb{R}}^{l,r}[\tilde{\sigma} + f_D](\mathbf{x}) \quad (3.17)$$

solve (2.5); moreover, for $\mathbf{x} \in \mathbb{R}^2 \cap \{|x_2| \leq L_C\}$, we can represent the solution by integrating along $\tilde{\Gamma}$

$$u^{l,r}(\mathbf{x}) = \mathcal{S}_{\tilde{\Gamma}}^{l,r}[\tilde{\tau} + f_N](\mathbf{x}) + \mathcal{D}_{\tilde{\Gamma}}^{l,r}[\tilde{\sigma} + f_D](\mathbf{x}). \quad (3.18)$$

For any sufficiently small $\epsilon > 0$, there exists an $0 < \eta \leq \epsilon$, and a constant C independent of ϵ such that the following holds. There is a unique $[\tilde{\sigma}_\epsilon, \tilde{\tau}_\epsilon] \in \mathcal{C}^{1/2, k} \oplus \mathcal{C}^{1, k}$ such that

$$\left(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}_\eta}\right) \begin{bmatrix} \tilde{\sigma}_\epsilon + f_D \\ \tilde{\tau}_\epsilon + f_N \end{bmatrix} (x_2) = \begin{bmatrix} f_D \\ f_N \end{bmatrix} (x_2) \quad \forall x_2 \in \tilde{\Gamma}_\eta. \quad (3.19)$$

Furthermore,

$$\left| \left(\mathcal{S}_{\tilde{\Gamma}_\eta}^{l,r}[\tilde{\tau}_\epsilon + f_N](\mathbf{x}) - \mathcal{D}_{\tilde{\Gamma}_\eta}^{l,r}[\tilde{\sigma}_\epsilon + f_D](\mathbf{x}) \right) - u^{l,r}(\mathbf{x}) \right| < C\epsilon \left(\|f_D\|_{\mathcal{C}_{\mathbb{R}}^{\frac{1}{2},0}} + \|f_N\|_{\mathcal{C}_{\mathbb{R}}^{1,0}} \right) \quad (3.20)$$

for all $\mathbf{x} \in \mathbb{R}^2 \cap \{|x_2| \leq L_C\}$.

4 Mapping Properties of operators

In this section we establish several properties of the operators $C_{\tilde{\Gamma}}$ and $D_{\tilde{\Gamma}}$. In particular, we show that the operators are independent of the choice of $\tilde{\Gamma} \in \mathcal{G}$ and are bounded between $\mathcal{C}^{\alpha,\beta}$ -spaces for appropriate choices of α and β .

We begin by defining the following convenient class of kernels.

Definition 7. Let $\alpha > 0$ and $\mathcal{A}_\alpha$ denote the class of kernels $\tilde{k}$ that have the following properties:

- The kernel $\tilde{k}$ is continuous on $(x_2, y_2) \in (\Gamma_{\mathbb{C}} \times \Gamma_{\mathbb{C}}) \setminus \{x_2 = y_2, |x_2| \leq d, |y_2| \leq d\}$.
- The kernel $\tilde{k}$ is weakly singular on the diagonal in $\Gamma_M \times \Gamma_M$, that is

$$\int_{\Gamma_M} |k(x_2, y_2)| dx_2 + \int_{\Gamma_M} |k(x_2, y_2)| dy_2 < \infty. \quad (4.1)$$

- The kernel $\tilde{k}$ is separately analytic in each variable, x_2, y_2 , in the interior of $\Gamma_{\mathbb{C}}$.
- There exist constants $C_{\pm, \pm}$ and K_1 such that

$$\left| \tilde{k}(x_2, y_2) - C_{\text{sign}(\Re x_2), \text{sign}(\Re y_2)} \frac{e^{ik(\text{sign}(\Re x_2)x_2 + \text{sign}(\Re y_2)y_2)}}{(\text{sign}(\Re x_2)x_2 + \text{sign}(\Re y_2)y_2)^\alpha} \right| \leq \frac{K_1 e^{-k(|\Im(x_2)| + |\Im(y_2)|)}}{(1 + |x_2| + |y_2|)^{\alpha + \frac{1}{2}}} \quad (4.2)$$

if $x_2, y_2 \in \Gamma_L \cup \Gamma_R$.

- There exists a constant K_2 and bounded continuous functions $f_\pm$ such that

$$\left| \tilde{k}(x_2, y_2) - f_{\text{sign}(\Re x_2)}(y_2) \frac{e^{ik \text{sign}(\Re x_2)x_2}}{(\text{sign}(\Re x_2)x_2)^\alpha} \right| \leq \frac{K_2 e^{-k|\Im(x_2)|}}{(1 + |x_2|)^{\alpha + \frac{1}{2}}} \quad (4.3)$$

if $x_2 \in \Gamma_L \cup \Gamma_R$ and $y_2 \in \Gamma_M$ and a similar statement holds with x_2 and y_2 swapped.

Remark 4. Note that both $\text{sign}(\Re x_2)x_2 + \text{sign}(\Re y_2)y_2$ and $\text{sign}(\Re x_2)x_2$ lie in the first quadrant, so their α^{th} -power is unambiguously defined by setting $z^\alpha = e^{\alpha \log z}$, where $\log z$ is defined in $\mathbb{C} \setminus (-\infty, 0]$ by setting $\log 1 = 0$.

The following theorem shows that the kernels k_C and k_D of the operators C and D can be analytically continued to $\Gamma_{\mathbb{C}} \times \Gamma_{\mathbb{C}}$ and lie in these kernel classes.

Theorem 4. The kernels k_C and k_D can be analytically continued to $\Gamma_{\mathbb{C}} \times \Gamma_{\mathbb{C}}$ and are in $\mathcal{A}_{\frac{3}{2}}$ and $\mathcal{A}_{\frac{1}{2}}$, respectively. Further, the kernel $k_C(x_2, y_2)$ has a $\log|x_2 - y_2|$ -singularity on the diagonal inside $[-d, d] \times [-d, d]$, and is continuously differentiable on the rest of $\Gamma_{\mathbb{C}} \times \Gamma_{\mathbb{C}}$. The kernel $k_D(x_2, y_2)$ has a $|x_2 - y_2|^2 \log|x_2 - y_2|$ -singularity in $[-d, d] \times [-d, d]$, and is continuously differentiable on the rest of $\Gamma_{\mathbb{C}} \times \Gamma_{\mathbb{C}}$.

Proof. The weak singularities of k_C , and k_D along the diagonal in $\Gamma_M \times \Gamma_M$ are derived in [20]. Proving that the kernels are analytic in $\Gamma_{\mathbb{C}}$ and that they satisfy (4.2) and (4.3) is straightforward, but tedious; and is left to Corollary 6 and Lemma 21 in Appendix B. $\square$

Remark 5. The remainder of the results in this section generalize to any pair of kernels belonging to $\mathcal{A}_{\frac{1}{2}}$ and $\mathcal{A}_{\frac{3}{2}}$. To obtain analogs of the main results, all that is required is a uniqueness result for (2.16) on the real line.

We begin by verifying that the complexified layer potentials exist when σ and τ are in the correct spaces.

Lemma 1. *For $0 < \alpha < \frac{1}{2}$ and $\beta > -k$, if $\sigma \in \mathcal{C}^{\alpha, \beta}$ and $\tau \in \mathcal{C}^{\alpha + \frac{1}{2}, \beta}$, then for any $x_2 \in \Gamma_{\mathbb{C}}$ and any $\tilde{\Gamma} \in \mathcal{G}$, the values of $C_{\tilde{\Gamma}}[\sigma](x_2)$ and $D_{\tilde{\Gamma}}[\tau](x_2)$ are given by convergent integrals.*

Proof. All of the kernels are locally integrable, so it is enough to verify that the integrand decays sufficiently rapidly at infinity. Since $k_C \in \mathcal{A}_{\frac{3}{2}}$ and $k_D \in \mathcal{A}_{\frac{1}{2}}$, the assumptions on σ and τ imply that the integrands of $C[\sigma](x_2)$ and $D[\tau](x_2)$ decay as $O(|y_2|^{-\frac{3}{2}-\alpha}e^{-(\beta+k)|\Im y_2|})$, and $O(|y_2|^{-1-\alpha}e^{-(\beta+k)|\Im y_2|})$ respectively. Since $\beta > -k$, these are both absolutely and uniformly integrable. $\square$

We now prove that $C_{\tilde{\Gamma}}$ and $D_{\tilde{\Gamma}}$ do not depend on the curve used to define them, which justifies denoting them simply by C and D .

Proposition 1. *Let $\sigma \in \mathcal{C}^{\alpha, \beta}$ and $\tau \in \mathcal{C}^{\alpha + \frac{1}{2}, \beta}$. For any $x_2 \in \Gamma_{\mathbb{C}}$, $C_{\tilde{\Gamma}}[\sigma](x_2)$ and $D_{\tilde{\Gamma}}[\tau](x_2)$ are independent of the choice $\tilde{\Gamma} \in \mathcal{G}$.*

Proof. Let $\tilde{\Gamma}$ and $\hat{\Gamma}$ be two curves in $\mathcal{G}$ and x_2 be any fixed point in $\Gamma_{\mathbb{C}}$. Let $\tilde{\Gamma}_{\rho}$ and $\hat{\Gamma}_{\rho}$ be the truncations of the contours $\tilde{\Gamma}$ and $\hat{\Gamma}$ at $|\Re y_2| = \rho$. Let $\gamma_{\rho}^{\pm}$ be the vertical lines connecting these truncations (see Figure 2). From the estimates in the proof of Lemma 1, it follows that

$$C_{\tilde{\Gamma}}[\sigma](x_2) = \lim_{\rho \rightarrow \infty} C_{\tilde{\Gamma}_{\rho}}[\sigma](x_2), \quad (4.4)$$

with a similar statement for $\hat{\Gamma}$. It is therefore enough to study the difference between $C_{\tilde{\Gamma}_{\rho}}$ and $C_{\hat{\Gamma}_{\rho}}$ and show that it vanishes as $\rho \rightarrow \infty$.

By Lemma 19 and the assumptions on (σ, τ) , both integrands are analytic in the region between $\tilde{\Gamma}$ and $\hat{\Gamma}$. We therefore can apply Cauchy's theorem to conclude that

$$C_{\tilde{\Gamma}_{\rho}}[\sigma](x_2) - C_{\hat{\Gamma}_{\rho}}[\sigma](x_2) = C_{\gamma_{\rho}}[\sigma](x_2) = \int_{\gamma_{\rho}^{+} \cup \gamma_{\rho}^{-}} k_C(x_2, y_2) \sigma(y_2) dy_2, \quad (4.5)$$

where $\gamma_{\rho}^{\pm}$ are oriented so that they start at $\hat{\Gamma}_{\rho}$ and end at $\tilde{\Gamma}_{\rho}$. We first consider γ_{ρ}^{+} ; since $-k < \beta$, the integral can be bounded by

$$|C_{\gamma_{\rho}^{+}}[\sigma](x_2)| \leq K \frac{\|\sigma\|_{\alpha, \beta}}{(\min_{\gamma_{\rho}^{+}} |y_2|)^{\alpha}} \int_0^{\infty} e^{-(\beta+k)t} dt \leq \frac{K}{\beta+k} \frac{\|\sigma\|_{\alpha, \beta}}{\rho^{\alpha}}. \quad (4.6)$$

As $\alpha > 0$, this bound implies that $|C_{\gamma_{\rho}^{+}}[\sigma](x_2)| \rightarrow 0$ as $\rho \rightarrow \infty$. A similar argument shows that $|C_{\gamma_{\rho}^{-}}[\sigma](x_2)| \rightarrow 0$ as $\rho \rightarrow \infty$ and thus have shown that $C_{\tilde{\Gamma}}[\sigma](x_2) = C_{\hat{\Gamma}}[\sigma](x_2)$ for any $x_2 \in \Gamma_{\mathbb{C}}$ and any pair of curves $\tilde{\Gamma}, \hat{\Gamma} \in \mathcal{G}$.

An essentially identical proof shows that $D_{\tilde{\Gamma}}[\tau](x_2)$ does not depend on $\tilde{\Gamma}$. $\square$

Since C and D are independent of the choice of $\tilde{\Gamma}$, in the sequel we suppress the dependence on the contour of integration $\tilde{\Gamma}$, except to emphasize quantities that explicitly depend on it. We now turn to proving that C and D are bounded on $\mathcal{C}^{\alpha, \beta}$. To do this, we need the following integral inequality; it is essentially Lemma 1 in [20].

Lemma 2. *If $0 \leq a < 1$ and $a + b > 1$, then*

$$I = \int_0^{\infty} \frac{1}{(|x_2| + t)^b} \frac{1}{(1+t)^a} dt \leq K |x_2|^{1-a-b}, \quad (4.7)$$

where K depends on a and b but not on $|x_2|$.

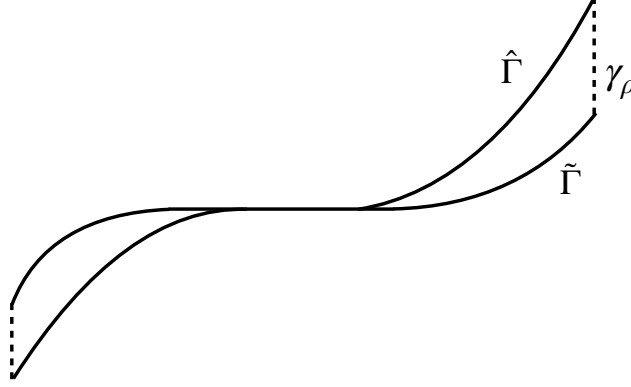


Figure 2: Contours for Proposition 1

Proof. We make the substitution $t = |x_2|s$ to get

$$I = \int_0^\infty \frac{1}{|x_2|^b(1+s)^b} \frac{1}{(1+|x_2|s)^a} |x_2| ds \leq \frac{1}{|x_2|^{a+b-1}} \int_0^\infty \frac{1}{(1+s)^b} \frac{1}{s^a} ds. \quad (4.8)$$

Since $0 < a < 1$, and $1 < a + b$ it is clear that this integral is finite. $\square$

In order to establish the decay of $C[\sigma]$ at infinity, we need to handle the contribution from inside the channel carefully owing to the logarithmic singularity of the kernel there [20]. In what follows it is helpful to introduce a partition of unity to isolate this singularity. Recall that $L_C > d$.

Definition 8. For $\epsilon = (L_C - d)/3$, let $\psi \in C^\infty$ be compactly supported on $[-L_C + \epsilon, L_C - \epsilon]$, identically one on $[-d - \epsilon, d + \epsilon]$, and $0 \leq \psi(y_2) \leq 1$ for all $y_2 \in \mathbb{R}$; set

$$k_{C_0}(x_2, y_2) = \psi(y_2)k_C(x_2, y_2) \quad \text{and} \quad k_{C_1} = k_C - k_{C_0}. \quad (4.9)$$

Let C_0 and C_1 be the integral operators with kernels k_{C_0} and k_{C_1} respectively.

The following lemma follows from Theorem 4.

Lemma 3. The kernel k_{C_0} is compactly supported in y_2 with a logarithmic singularity on the diagonal.

$$|k_{C_0}(x_2, y_2)| \leq K\psi(y_2) \frac{[1 + \psi(x_2)|\log|x_2 - y_2||]e^{-k|\Im x_2|}}{(1 + |x_2|)^{\frac{3}{2}}}. \quad (4.10)$$

Using this split of the kernel of C , and Lemma 2, we establish the boundedness of the operators $\sigma \mapsto C[\sigma]$ and $\tau \mapsto D[\tau]$ between appropriate $\mathcal{C}^{\alpha, \beta}$ spaces in the following three lemmas. We begin by showing that $C[\sigma]$ and $D[\tau]$ have the required continuity and analyticity.

Lemma 4. If $\sigma \in \mathcal{C}^{\alpha, \beta}$ and $\tau \in \mathcal{C}^{\alpha + \frac{1}{2}, \beta}$, then $C[\sigma]$ and $D[\tau]$ are continuous on $\Gamma_{\mathbb{C}}$.

Proof. Since $k_{C_1}(x_2, y_2)\sigma(y_2)$ and $k_D(x_2, y_2)\tau(y_2)$ are continuous and can be bounded independent of x_2 , the functions $C_1[\sigma]$ and $D[\tau]$ are clearly continuous by the dominated convergence theorem. The kernel k_{C_0} is weakly singular and supported on $|y_2| < L_C$. Standard arguments for compact domains (e.g. Theorem 1.11 in [14]) therefore give that $C_0[\sigma]$ is continuous. $\square$

Lemma 5. If $\sigma \in \mathcal{C}^{\alpha, \beta}$ and $\tau \in \mathcal{C}^{\alpha + \frac{1}{2}, \beta}$, then $C[\sigma](x_2)$ and $D[\tau](x_2)$ are analytic in the interior of $\Gamma_{\mathbb{C}}$.

Proof. The proof follows from Morera's theorem. Let γ be a closed contour in the first quadrant. Since, $k_C \in \mathcal{A}_{\frac{3}{2}}$, $\sigma \in \mathcal{C}^{\alpha,\beta}$, and γ is bounded we can use Fubini's theorem to see that

$$\int_{\gamma} C[\sigma](x_2) dx_2 = \int_{\tilde{\Gamma}} \int_{\gamma} k_C(x_2, y_2) \sigma(y_2) dx_2 dy_2. \quad (4.11)$$

The inner integral is zero because $k_C(x_2, y_2)$ is an analytic function of x_2 and so we have that $C[\sigma]$ is analytic. An identical proof can be used to show that $D[\tau]$ is also analytic. $\square$

Remark 6. From the proof of this theorem it is clear that it is not necessary for σ or τ to be analytic. Given a curve $\tilde{\Gamma} \in \mathcal{G}$, all that is required for $C_{\tilde{\Gamma}}[\sigma](x_2)$, $D_{\tilde{\Gamma}}[\tau](x_2)$ to have analytic continuations to $\Gamma_L \cup \Gamma_R$ is for the integrals defining $C_{\tilde{\Gamma}}[\sigma](x_2)$ and $D_{\tilde{\Gamma}}[\tau](x_2)$ to be absolutely convergent, uniformly for x_2 in compact subsets of $\text{int } \Gamma_L \cup \Gamma_R$. This property holds for $\sigma \in \mathcal{C}_{\tilde{\Gamma}}^{\alpha,\beta}$ and $\tau \in \mathcal{C}_{\tilde{\Gamma}}^{\alpha+\frac{1}{2},\beta}$ respectively, but one cannot generally change the contour of integration in the y_2 -integral, unless the data have analytic continuations from $\tilde{\Gamma}$.

Finally, we prove the norm estimates for the operators C and D in the lemma below.

Lemma 6. Suppose that $\sigma \in \mathcal{C}^{\alpha,\beta}$ and $\tau \in \mathcal{C}^{\alpha+\frac{1}{2},\beta}$, with $0 < \alpha < 1/2$ and $\beta > -k$. Then there exists a $K > 0$ such that

$$\|C[\sigma]\|_{\alpha+\frac{1}{2},k} \leq K \|\sigma\|_{\alpha,\beta} \quad (4.12)$$

and

$$\|D[\tau]\|_{\alpha,k} \leq K \|\tau\|_{\alpha+\frac{1}{2},\beta}. \quad (4.13)$$

Proof. We prove this bound using the choice $\tilde{\Gamma} = \mathbb{R}$. We begin by proving the bound for

$$C_1[\sigma](x_2) = \int_{-\infty}^{\infty} k_{C_1}(x_2, t) \sigma(t) dt. \quad (4.14)$$

For $t \in \mathbb{R}$, $|\sigma(t)| \leq \|\sigma\|_{\alpha,\beta} / (1 + |t|)^{\alpha}$. Using this bound, and Theorem 4, we have that

$$|C_1[\sigma](x_2)| \leq \int_0^{\infty} \frac{K e^{-k|\Im x_2|}}{(1 + |x_2| + t)^{\frac{3}{2}}} \frac{\|\sigma\|_{\alpha,\beta}}{(1 + t)^{\alpha}} dt. \quad (4.15)$$

If we factor out the terms independent of t , then Lemma 2 with $a = \alpha$ and $b = \frac{3}{2}$ gives

$$|C_1[\sigma](x_2)| \leq \frac{K' e^{-k\Im x_2}}{(1 + |x_2|)^{\frac{1}{2} + \alpha}} \|\sigma\|_{\alpha,\beta}. \quad (4.16)$$

The bound on $|C_1[\sigma](x_2)|$ for $|x_2| < 1$ is obvious, and therefore we see that there is a constant C so that $\|C_1[\sigma]\|_{\alpha+\frac{1}{2},k} \leq C \|\sigma\|_{\alpha,\beta}$.

To bound the contribution of C_0 , we note that

$$|C_0[\sigma](x_2)| \leq \int_{-L_C}^{L_C} \frac{2K |\log |x_2 - t|| e^{-k|\Im x_2|}}{(1 + |x_2|)^{\frac{3}{2}}} \|\sigma\|_{\alpha,\beta} dt. \quad (4.17)$$

Since the integral is over a finite domain, we can bound the integral of the logarithm by $C\sqrt{|x_2|}$ to see

$$|C_0[\sigma](x_2)| \leq \frac{K' e^{-k|\Im x_2|}}{1 + |x_2|} \|\sigma\|_{\alpha,\beta}, \quad (4.18)$$

which gives (4.12).

A similar argument to the one above gives

$$|D[\tau](x_2)| \leq \int_0^{\infty} \frac{K e^{-k|\Im x_2|}}{(1 + |x_2| + t)^{\frac{1}{2}}} \frac{\|\tau\|_{\alpha+\frac{1}{2},\beta}}{(1 + t)^{\alpha+\frac{1}{2}}} dt. \quad (4.19)$$

Applying Lemma 2 with $a = \alpha + \frac{1}{2}$ and $b = \frac{1}{2}$ gives (4.13). $\square$

Remark 7. By carrying out the estimate on another choice of contour $\tilde{\Gamma} \in \mathcal{G}$, it is clear that the conclusion of Lemma 6 actually holds for any such curve and functions $\sigma_{\tilde{\Gamma}} \in \mathcal{C}_{\tilde{\Gamma}}^{\alpha,\beta}$ and $\tau_{\tilde{\Gamma}} \in \mathcal{C}_{\tilde{\Gamma}}^{\alpha+\frac{1}{2},\beta}$, in that there exists a $K > 0$ such that,

$$\|C_{\tilde{\Gamma}}[\sigma_{\tilde{\Gamma}}]\|_{\alpha+\frac{1}{2},k} \leq K \|\sigma_{\tilde{\Gamma}}\|_{\alpha,\beta,\tilde{\Gamma}} \quad (4.20)$$

and

$$\|D_{\tilde{\Gamma}}[\tau_{\tilde{\Gamma}}]\|_{\alpha,k} \leq K \|\tau_{\tilde{\Gamma}}\|_{\alpha+\frac{1}{2},\beta,\tilde{\Gamma}}. \quad (4.21)$$

Here $C_{\tilde{\Gamma}}[\sigma_{\tilde{\Gamma}}]$, $D_{\tilde{\Gamma}}[\tau_{\tilde{\Gamma}}]$ are the analytic continuations described in Remark 6.

Combining the above lemmas gives the following theorem which proves the boundedness of C and D between appropriate $\mathcal{C}^{\alpha,\beta}$ spaces.

Theorem 5. For any $0 < \alpha < \frac{1}{2}$ and $-k < \beta$, the following maps are bounded

$$C : \mathcal{C}^{\alpha,\beta} \rightarrow \mathcal{C}^{\alpha+\frac{1}{2},k} \quad \text{and} \quad D : \mathcal{C}^{\alpha+\frac{1}{2},\beta} \rightarrow \mathcal{C}^{\alpha,k}. \quad (4.22)$$

Since $\mathcal{C}^{\alpha',k}$ can be continuously embedded into $\mathcal{C}^{\alpha',\beta}$ for any $\alpha' > 0$ and $\beta \leq k$, this theorem also implies the weaker statement that

$$C : \mathcal{C}^{\alpha,\beta} \rightarrow \mathcal{C}^{\alpha+\frac{1}{2},\beta} \quad \text{and} \quad D : \mathcal{C}^{\alpha+\frac{1}{2},\beta} \rightarrow \mathcal{C}^{\alpha,\beta} \quad (4.23)$$

are bounded operators for $0 < \alpha < \frac{1}{2}$ and $-k < \beta \leq k$.

5 Analysis of the integral equation system

We now turn our attention to the analysis of (3.8). We begin by proving uniqueness, and then show that the operator $\mathcal{I} + \mathcal{K}$ is a Fredholm operator of index zero. This combined with uniqueness implies that (3.8) is solvable for arbitrary data in $\mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$, with $0 < \alpha < \frac{1}{2}$, $-k < \beta \leq k$.

5.1 Uniqueness

The uniqueness of the solution to (3.8) is essentially immediate from the uniqueness result for $(\mathcal{I} + \mathcal{K}_{\mathbb{R}})$ proved in [23], and analyticity.

Lemma 7 (Uniqueness). For $0 < \alpha < \frac{1}{2}$, $-k < \beta \leq k$, if $[f_D, f_N] \in \mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$, then there is at most one solution $[\sigma, \tau] \in \mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$ of (3.8).

Proof. Since the equation is linear, we need to show that the only solution $[\sigma, \tau] \in \mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$ to

$$(\mathcal{I} + \mathcal{K}) \begin{bmatrix} \sigma \\ \tau \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (5.1)$$

is $\sigma = \tau = 0$. From Proposition 1 it follows that if $[\sigma, \tau]$ solves (5.1), then $(\mathcal{I} + \mathcal{K}_{\mathbb{R}}) \begin{bmatrix} \sigma \\ \tau \end{bmatrix} (x_2) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, for $x_2 \in \mathbb{R}$.

The restriction $[\sigma, \tau]_{\mathbb{R}} \in \mathcal{C}^{\alpha}(\mathbb{R}) \oplus \mathcal{C}^{\alpha+\frac{1}{2}}(\mathbb{R})$ and therefore Theorem 3 in [23] shows that $[\sigma, \tau]_{\mathbb{R}} = [0, 0]$. The proof then follows from (3.9) with $\tilde{\Gamma} = \mathbb{R}$. $\square$

This result has the following useful corollary

Corollary 2. If, for any $\tilde{\Gamma} \in \mathcal{G}$, there is a pair $[\sigma_{\tilde{\Gamma}}, \tau_{\tilde{\Gamma}}] \in \mathcal{C}_{\tilde{\Gamma}}^{\alpha,\beta} \oplus \mathcal{C}_{\tilde{\Gamma}}^{\alpha+\frac{1}{2},\beta}$ such that

$$(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}}) \begin{bmatrix} \sigma_{\tilde{\Gamma}} \\ \tau_{\tilde{\Gamma}} \end{bmatrix} (x_2) = 0 \text{ for } x_2 \in \tilde{\Gamma}, \quad (5.2)$$

then $[\sigma_{\tilde{\Gamma}}, \tau_{\tilde{\Gamma}}]$ has an analytic continuation to an element $[\sigma, \tau] \in \mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$ and $[\sigma, \tau] \equiv [0, 0]$.

Proof. Let $[\sigma_{\tilde{\Gamma}}, \tau_{\tilde{\Gamma}}]$ be a solution to (5.2). As noted in Remark 7, the components of $\mathcal{K}_{\tilde{\Gamma}} \begin{bmatrix} \sigma_{\tilde{\Gamma}} \\ \tau_{\tilde{\Gamma}} \end{bmatrix}$ extend analytically from $\tilde{\Gamma}$ to $\Gamma_L \cup \Gamma_R$, to define an element of $\mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}$. Since

$$\begin{bmatrix} \sigma_{\tilde{\Gamma}} \\ \tau_{\tilde{\Gamma}} \end{bmatrix} (x_2) = -\mathcal{K}_{\tilde{\Gamma}} \begin{bmatrix} \sigma_{\tilde{\Gamma}} \\ \tau_{\tilde{\Gamma}} \end{bmatrix} (x_2), \quad (5.3)$$

for $x_2 \in \tilde{\Gamma}$, it follows that $[\sigma_{\tilde{\Gamma}}, \tau_{\tilde{\Gamma}}]$ also has an analytic continuation belonging to $\mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}$. This pair satisfies

$$(\mathcal{I} + \mathcal{K}) \begin{bmatrix} \sigma \\ \tau \end{bmatrix} (x_2) = 0 \text{ for } x_2 \in \Gamma_{\mathbb{C}}. \quad (5.4)$$

Lemma 7 implies that $[\sigma, \tau] \equiv [0, 0]$. $\square$

5.2 The Fredholm Property and Existence of Solutions

We now show that the operator defining the integral equation (3.8) is Fredholm of index 0. The argument is nearly identical to the discussion of the Fredholm property of (2.16) in [20], but is given here for completeness.

For integral equations on bounded domains, compactness is usually proved by noting that the operators are smoothing and using the Arzela-Ascoli theorem. Since the Arzela-Ascoli theorem is only applicable on bounded domains, this proof cannot be directly generalized to our situation. Instead, we must prove that the integral operator of interest, in addition to smoothing, maps $\mathcal{C}^{\alpha, \beta}$ into a space of functions with faster decay.

Proposition 2. *Suppose $0 < \alpha < \alpha'$ and $\beta \in \mathbb{R}$. Let $A : \mathcal{C}^{\alpha, \beta} \rightarrow \mathcal{C}^{\alpha', \beta}$ be a bounded linear operator such that $\partial_{x_2} A : \mathcal{C}^{\alpha, \beta} \rightarrow \mathcal{C}^{0, \beta}$ boundedly. Then $A : \mathcal{C}^{\alpha, \beta} \rightarrow \mathcal{C}^{\alpha, \beta}$ is a compact operator.*

Proof. Let $\{f_n\}$ be a bounded sequence in $\mathcal{C}^{\alpha, \beta}$. Since $\{Af_n\}$ is bounded in $\mathcal{C}^{\alpha', \beta}$ and $\alpha < \alpha'$, for any $\varepsilon > 0$ there exists a bounded subset B_ε of $\Gamma_{\mathbb{C}}$ such that

$$\|Af_n\|_{\mathcal{C}^{\alpha, \beta}(\Gamma_{\mathbb{C}} \setminus B_\varepsilon)} < \varepsilon \quad (5.5)$$

for all n . Since Af_n and $(\partial_{x_2} Af_n)$ are uniformly bounded in $\mathcal{C}^{0, \beta}$, the set $\{Af_n\}$ is uniformly bounded and uniformly equi-continuous on $B_\varepsilon \cap \Gamma_{\mathbb{C}}$. By the Arzela-Ascoli theorem, there exists a Cauchy subsequence of Af_n in $\mathcal{C}^0(B_\varepsilon \cap \Gamma_{\mathbb{C}})$, and so also in $\mathcal{C}^{\alpha, \beta}(\Gamma_{\mathbb{C}} \cap B_\varepsilon)$. By repeating this for $\varepsilon \in \{\frac{1}{m} : m \in \mathbb{N}\}$ we can use (5.5) in a standard diagonalization argument to construct a subsequence of $\{Af_n\}$ that converges in $\mathcal{C}^{\alpha, \beta}$. As this holds for all bounded sequences, it follows that $A : \mathcal{C}^{\alpha, \beta} \rightarrow \mathcal{C}^{\alpha, \beta}$ is a compact operator. $\square$

Unfortunately, this proposition cannot be applied directly to $\mathcal{K}$, since D does not increase the algebraic rate of decay. Instead, we follow [20] and observe that

$$\begin{bmatrix} \mathcal{I} & -D \\ 0 & \mathcal{I} \end{bmatrix} \begin{bmatrix} \mathcal{I} & D \\ C & \mathcal{I} \end{bmatrix} \begin{bmatrix} \mathcal{I} & 0 \\ -C & \mathcal{I} \end{bmatrix} = \begin{bmatrix} \mathcal{I} - DC & 0 \\ 0 & \mathcal{I} \end{bmatrix}. \quad (5.6)$$

Since the left and right factors on the left-hand side of (5.6) are bounded and invertible, equation (3.8) is Fredholm of index 0 if and only if the right hand side of (5.6) is. In what follows, we prove that DC is a compact map from $\mathcal{C}^{\alpha, \beta}$ to itself. The advantage of using the composition is that the oscillations of the kernels cause the kernel of the composite map DC to decay faster than is predicted by Theorem 5 and allow us to use Proposition 2.

To aid in the proof, we split the kernels into far field and near field components. Specifically, let ψ be the bump function specified in Definition 8 and set

$$k_{C_f}(x_2, y_2) = (1 - \psi(x_2))(1 - \psi(y_2))k_C(x_2, y_2) \quad \text{and} \quad k_{C_n}(x_2, y_2) = k_C(x_2, y_2) - k_{C_f}(x_2, y_2). \quad (5.7)$$

we define k_{D_n} and k_{D_f} similarly. This splitting ensures that k_{C_f} and k_{D_f} are zero if either x_2 or y_2 is in the waveguide, which gives them a simple asymptotic form. The operators with kernels $k_{C_n}, k_{C_f}, k_{D_n}$, and k_{D_f} are denoted C_n, C_f, D_n and D_f , respectively. Using these definitions, we can split the product DC :

$$DC = D_n C_n + D_f C_n + D_n C_f + D_f C_f \quad (5.8)$$

The proof of Theorem 5 shows that all of these are bounded from $\mathcal{C}^{\alpha, \beta}$ to itself.

We now prove that C_n and D_n are compact.

Lemma 8. *The operator $\begin{bmatrix} 0 & D_n \\ C_n & 0 \end{bmatrix}$ is compact from $\mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}$ to $\mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}$*

Proof. We expand the definition of k_{C_n} :

$$k_{C_n}(x_2, y_2) = \psi(x_2)k_C(x_2, y_2) + \psi(y_2)k_C(x_2, y_2) - \psi(x_2)\psi(y_2)k_C(x_2, y_2). \quad (5.9)$$

The first and third terms define smoothing integral operators, mapping into functions which are compactly supported on Γ_M . They therefore trivially satisfy the assumptions of Proposition 2 and so are compact from $\mathcal{C}^{\alpha, \beta}$ to $\mathcal{C}^{\alpha + \frac{1}{2}, \beta}$.

For the second term, we let

$$\tilde{k}_{C_n}(x_2, y_2) = (1 - \psi(x_2)) \frac{e^{ik \operatorname{sign}(\Re x_2)x_2}}{(\operatorname{sign}(\Re x_2)x_2)^{3/2}} f_{C, \operatorname{sign}(\Re x_2)}(y_2) \psi(y_2) \quad (5.10)$$

be the asymptotic form of $k_C(x_2, y_2)\psi(y_2)$ identified by Theorem 4. Theorem 4 guarantees that

$$|k_C(x_2, y_2)\psi(y_2) - \tilde{k}_{C_n}(x_2, y_2)| \leq \frac{K e^{-k|\Im x_2|}}{|x_2|^2} \psi(y_2). \quad (5.11)$$

Since $k_C(x_2, y_2)\psi(y_2) - \tilde{k}_{C_n}(x_2, y_2)$ decays more rapidly than k_C and has decaying derivatives, it defines a compact operator from $\mathcal{C}^{\alpha, \beta}$ to $\mathcal{C}^{\alpha + 1/2, \beta}$ by Proposition 2 and a proof similar to that of Lemma 6. The operator with kernel $\tilde{k}_{C_n}(x_2, y_2)$, being rank one and bounded, is also a compact operator. Moreover, it trivially maps functions in $\mathcal{C}^{\alpha, \beta}$ to $\mathcal{C}^{\alpha + 1/2, \beta}$, for any $\alpha \leq 1$, and $\beta \leq k$. The operator defined by $k_C(x_2, y_2)\psi(y_2)$ is thus the sum of a compact operator and a rank one operator and so compact from $\mathcal{C}^{\alpha, \beta}$ to $\mathcal{C}^{\alpha + 1/2, \beta}$.

The operator D_n can be seen to be compact using a nearly identical proof. $\square$

Since the operators C_f, C_n, D_f, D_n are bounded, these lemmas show that all terms in (5.8) are compact, but for $D_f C_f$, which we now consider.

Lemma 9. *The operator $D_f C_f$ is compact from $\mathcal{C}^{\alpha, \beta}$ to itself.*

Proof. We showed previously that, for $|x_2| > d$ and $|y_2| > d$ the kernels k_D and k_C have leading order asymptotics:

$$\tilde{k}_D(x_2, y_2) = \frac{C_{\pm, \pm} e^{ik(\pm x_2 \pm y_2)}}{(\pm x_2 \pm y_2)^{\frac{1}{2}}} \quad \text{and} \quad \tilde{k}_C(x_2, y_2) = \frac{C'_{\pm, \pm} e^{ik(\pm x_2 \pm y_2)}}{(\pm x_2 \pm y_2)^{\frac{3}{2}}}. \quad (5.12)$$

where the choice of $\pm$ depends on the signs of $\Re x_2$ and $\Re y_2$. In analogy with above, we can define the far field components of $\tilde{k}_D$ and $\tilde{k}_C$, which we denote by $\tilde{k}_{D_f}$ and $\tilde{k}_{C_f}$, respectively. By Theorem 4,

$$|k_{D_f}(x_2, y_2) - \tilde{k}_{D_f}(x_2, y_2)| \leq \frac{K e^{-k(|\Im x_2| + |\Im y_2|)}}{|x_2| + |y_2|}. \quad (5.13)$$

We have a similar expression for k_{C_f} , except that the denominator is $(|x_2| + |y_2|)^2$. Since $k_{D_f} - \tilde{k}_{D_f}$ and $k_{C_f} - \tilde{k}_{C_f}$ decay more rapidly and have decaying derivatives, a proof similar to the proofs of Lemma 6 and Proposition 2 shows that they define compact operators.

We now study the behavior of $\tilde{D}_f \tilde{C}_f$, the product of the operators defined by the asymptotic kernels. We denote the kernel of this operator by $\tilde{k}$. For notational convenience, in the following we set $\psi^c(z) = 1 - \psi(z)$. To bound $\tilde{k}$, we suppose that $x_2, y_2 \in \Gamma_R$ and integrate over the real line

$$\tilde{k}(x_2, y_2) = \int_{-\infty}^{\infty} \tilde{k}_{D_f}(x_2, z_2) \tilde{k}_{C_f}(z_2, y_2) dz_2. \quad (5.14)$$

We split this integral into the regions $\{z_2 > d\}$ and $\{z_2 < -d\}$, and first consider the integral over $[d, \infty)$,

$$\tilde{k}_+(x_2, y_2) = C \int_d^{\infty} \frac{e^{ik(x_2+z_2)}}{(x_2+z_2)^{\frac{1}{2}}} \frac{\psi^c(z_2) e^{ik(z_2+y_2)}}{(z_2+y_2)^{\frac{3}{2}}} dz_2 = C e^{ik(x_2+y_2)} \int_d^{\infty} \frac{\psi^c(z_2) e^{2ikz_2}}{(x_2+z_2)^{\frac{1}{2}} (z_2+y_2)^{\frac{3}{2}}} dz_2. \quad (5.15)$$

We need to estimate the integral

$$I = \int_d^{\infty} \frac{\psi^c(z_2) e^{2ikz_2}}{(x_2+z_2)^{\frac{1}{2}} (z_2+y_2)^{\frac{3}{2}}} dz_2. \quad (5.16)$$

If we integrate by parts twice, we find that

$$\begin{aligned} I = & -\frac{1}{4k^2} \int_d^{\infty} \psi^c(z_2) e^{2ikz_2} \left[\frac{15}{4(x_2+z_2)^{\frac{1}{2}} (z_2+y_2)^{7/2}} \right. \\ & + \frac{3}{2(x_2+z_2)^{\frac{3}{2}} (z_2+y_2)^{\frac{5}{2}}} + \left. \frac{3}{4(x_2+z_2)^{\frac{5}{2}} (z_2+y_2)^{\frac{3}{2}}} \right] dz_2 \\ & + \frac{1}{4k^2} \int_d^{\infty} \psi^{c'}(z_2) e^{2ikz_2} \left[\frac{3x_2+y_2+4z_2}{2(x_2+z_2)^{\frac{3}{2}} (z_2+y_2)^{\frac{5}{2}}} \right] dz_2 \\ & - \frac{1}{4k^2} \int_d^{\infty} \psi^{c''}(z_2) e^{2ikz_2} \frac{1}{(x_2+z_2)^{\frac{1}{2}} (z_2+y_2)^{\frac{3}{2}}} dz_2 \end{aligned} \quad (5.17)$$

Observing that $\psi^{c'}$ and $\psi^{c''}$ are compactly supported in $[-L_C, -d] \cup [d, L_C]$, each of these terms is easily seen to be bounded in absolute value by $\frac{C}{(|x_2+d|)^{\frac{1}{2}} (|y_2+d|)^{\frac{3}{2}}}$, for some appropriately chosen constant C . Plugging this into our expression for $\tilde{k}$ gives

$$|\tilde{k}_+(x_2, y_2)| \leq \frac{C e^{-k\Im(x_2+y_2)}}{(|x_2+d|)^{\frac{1}{2}} (|y_2+d|)^{\frac{3}{2}}}. \quad (5.18)$$

A similar proof can be applied to bound the integral over the region $\{z_2 < -d\}$.

Repeating the derivation for x_2 and y_2 in different regions of $\Gamma_L \cup \Gamma_R$ then gives

$$|\tilde{k}(x_2, y_2)| \leq \frac{C e^{-k(|\Im x_2| + |\Im y_2|)}}{(|\pm x_2 + d|)^{\frac{1}{2}} (|\pm y_2 + d|)^{\frac{3}{2}}}. \quad (5.19)$$

A similar calculation shows that,

$$|\partial_{x_2} \tilde{k}(x_2, y_2)| \leq \frac{C e^{-k(|\Im x_2| + |\Im y_2|)}}{(|\pm x_2 + d|)^{\frac{3}{2}} (|\pm y_2 + d|)^{\frac{3}{2}}}. \quad (5.20)$$

If we repeat the proof from Section 4, then these estimates are enough to prove that

$$\tilde{D}_f \tilde{C}_f : \mathcal{C}^{\alpha, \beta} \rightarrow \mathcal{C}^{\frac{1}{2}, \beta} \quad \text{and} \quad \partial_{x_2} \tilde{D}_f \tilde{C}_f : \mathcal{C}^{\alpha, \beta} \rightarrow \mathcal{C}^{\frac{3}{2}, \beta}. \quad (5.21)$$

By Proposition 2 we thus have that $\tilde{D}_f \tilde{C}_f$ is a compact map from $\mathcal{C}^{\alpha, \beta}$ to itself. Thus

$$D_f C_f = \tilde{D}_f \tilde{C}_f + (D_f - \tilde{D}_f) \tilde{C}_f + \tilde{D}_f (C_f - \tilde{C}_f) + (D_f - \tilde{D}_f) (C_f - \tilde{C}_f) \quad (5.22)$$

is compact. $\square$

Assembling the above results gives the following theorem.

Theorem 6. For $0 < \alpha < \frac{1}{2}$, $-k < \beta \leq k$, the operator DC is a compact operator from $\mathcal{C}^{\alpha,\beta}$ to itself. Equation (5.6) is a second kind Fredholm integral equation, and therefore of index 0. Equation (3.8) is therefore also Fredholm with index zero.

Coupled with the uniqueness result, Lemma 7, we have the following basic existence result.

Corollary 3 (Existence). For $0 < \alpha < \frac{1}{2}$, $-k < \beta \leq k$, given $[f_D, f_N] \in \mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$ the equation $(\mathcal{I} + \mathcal{K}) \begin{bmatrix} \sigma \\ \tau \end{bmatrix} = \begin{bmatrix} f_D \\ f_N \end{bmatrix}$ has a unique solution in $\mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$.

By restriction this implies an existence result on any $\tilde{\Gamma} \in \mathcal{G}$.

Lemma 10. Let $\tilde{\Gamma}$ be any curve in $\mathcal{G}$. If $[f_D, f_N] \in \mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$, then there exists a solution of $[\sigma, \tau] \in \mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$ of $(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}}) \begin{bmatrix} \sigma \\ \tau \end{bmatrix} = \begin{bmatrix} f_D \\ f_N \end{bmatrix}_{\tilde{\Gamma}}$

Proof. The solution of the equation on $\tilde{\Gamma}$ is simply the solution given in the corollary restricted to $\tilde{\Gamma}$. $\square$

Theorem 1 then follows from the corollary and Lemma 10. We also observe the following fact about the relationship between any solution of equation (3.8) on a curve $\tilde{\Gamma} \in \mathcal{G}$, and the analytic extension of the solution $[\sigma, \tau]$, given data $[f_D, f_N] \in \mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$. This is an extension of the argument used to prove Corollary 2.

Lemma 11. If $[f_D, f_N] \in \mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$, and $[\sigma_{\tilde{\Gamma}}, \tau_{\tilde{\Gamma}}] \in \mathcal{C}_{\tilde{\Gamma}}^{\alpha,\beta} \oplus \mathcal{C}_{\tilde{\Gamma}}^{\alpha+\frac{1}{2},\beta}$ solves

$$(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}}) \begin{bmatrix} \sigma_{\tilde{\Gamma}} \\ \tau_{\tilde{\Gamma}} \end{bmatrix} = \begin{bmatrix} f_D \\ f_N \end{bmatrix}_{\tilde{\Gamma}}, \quad (5.23)$$

then $[\sigma_{\tilde{\Gamma}}, \tau_{\tilde{\Gamma}}]$ extends to $\Gamma_{\mathbb{C}}$, where it defines an element of $\mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$ that solves (3.8), and hence agrees with the solution found in Lemma 10.

Proof. Rewriting the equation we see that, for $x_2 \in \tilde{\Gamma}$,

$$\begin{bmatrix} \sigma_{\tilde{\Gamma}} \\ \tau_{\tilde{\Gamma}} \end{bmatrix} (x_2) = \begin{bmatrix} f_D \\ f_N \end{bmatrix} (x_2) - \mathcal{K}_{\tilde{\Gamma}} \begin{bmatrix} \sigma_{\tilde{\Gamma}} \\ \tau_{\tilde{\Gamma}} \end{bmatrix} (x_2). \quad (5.24)$$

From the hypothesis of the lemma, and Remarks 6 and 7, both terms on the right hand side have analytic extensions to $x_2 \in \Gamma_{\mathbb{C}}$, which define an element of $[\sigma, \tau] \in \mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$, analytically extending $[\sigma_{\tilde{\Gamma}}, \tau_{\tilde{\Gamma}}]$. The lemma is immediate from this observation. $\square$

Remark 8. Taken together Corollary 2, Lemma 10, and Lemma 11 show that, for data in $\mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$, the problem of solving equation (2.16) on the real line, with appropriate estimates, can be replaced with that of solving (5.23) for a convenient choice of contour, $\tilde{\Gamma}$. This is the basis for the complexification technique that we employ for our numerical examples. Note that the solution to the latter problem restricted to $\tilde{\Gamma} \cap \mathbb{R}$ agrees with the solution to (2.16) on this subset.

In fact, the arguments presented above can be used to show that $(\mathcal{I} - D_{\tilde{\Gamma}} C_{\tilde{\Gamma}}) : \mathcal{C}_{\tilde{\Gamma}}^{\alpha,\beta} \rightarrow \mathcal{C}_{\tilde{\Gamma}}^{\alpha,\beta}$ is Fredholm of index 0. Corollary 2 then shows that this operator has a trivial null-space and is therefore invertible.

6 Solution to PDE (1.1)

We now check that the solution of (3.8) gives a solution of the transmission problem (2.5). This is proved in [20] for data in $\mathcal{C}^\alpha(\mathbb{R}) \oplus \mathcal{C}^{\alpha+\frac{1}{2}}(\mathbb{R})$, for $0 < \alpha < \frac{1}{2}$. If the data also satisfies the outgoing conditions in (3.13), then the solution to the transmission problem and PDE satisfies the outgoing radiation condition given in [23]. In this section we consider the extent to which the representations of $u^{l,r}(\mathbf{x})$, for $\mathbf{x} \in \mathbb{R}^2$, as an integral over the interface $\{(0, x_2) \mid x_2 \in \mathbb{R}\}$ can be replaced with integrals over a curve $\tilde{\Gamma} \in \mathcal{G}$, assuming the data belongs to $\mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$. Because the analytic continuations of the fundamental solutions of the bi-infinite waveguide problems do not necessarily decay at a uniform exponential rate along curves in $\mathcal{G}$, in order to compute $u^{l,r}$ using our representation (2.9), we impose the additional restriction $0 \leq \beta$.

To deform the contour of integration, we consider the single and double layer contributions separately. The Green's functions are written as $G^{l,r}(\mathbf{x}; \mathbf{y}) = G_0(\mathbf{x}; \mathbf{y}) + w^{l,r}(\mathbf{x}; \mathbf{y})$; as the two terms have rather different analytic continuation properties, we treat them separately. Since the kernels $w^{l,r}(x_1, x_2; y_1, y_2)$ depend only on $x_1 - y_1$, it is convenient to represent them in terms of their partial Fourier transform, i.e.

$$w^{l,r}(x_1, x_2; y_1, y_2) = \int_{\gamma_F} e^{i\xi(x_1-y_1)} \tilde{w}^{l,r}(\xi; x_2, y_2) d\xi, \quad (6.1)$$

where γ_F is defined to be $\{x + iy : y = -\tanh(x)\}$. In every bounded subset of $\mathbb{R}^2 \times \mathbb{R}^2$, this representation is the same as the one presented in [20], see Appendix A for a discussion of the properties of $\tilde{w}^{l,r}(\xi; x_2, y_2)$.

As shown in Lemma 19, the analytic continuations of the kernels $w^{l,r}$ have the same asymptotic properties when $x_1 \neq 0$ as they do when $x_1 = 0$. Hence these terms can be analytically continued to any curve in $\mathcal{G}$, and we easily show that, for any $(x_1, x_2) \in \mathbb{R}^2$, $\tau \in \mathcal{C}^{\alpha+\frac{1}{2},\beta}$, and $\tilde{\Gamma} \in \mathcal{G}$

$$\int_{\mathbb{R}} w^{l,r}(x_1, x_2; 0, y_2) \tau(y_2) dy_2 = \int_{\tilde{\Gamma}} w^{l,r}(x_1, x_2; 0, y_2) \tau(y_2) dy_2, \quad (6.2)$$

with a similar result for the σ -integral.

What requires more care is the analytic continuation of the free-space contributions. This is because the free-space kernel on $\{y_1 = 0\}$ is a function of $\sqrt{x_1^2 + (y_2 - x_2)^2}$, so care must be taken to avoid branch cuts of this function as y_2 moves into the complex plane. For stability it is also important to keep $\Im \sqrt{x_1^2 + (y_2 - x_2)^2} \geq 0$. If we require that $x_2 \in [-L_C, L_C]$, then these conditions hold for $y_2 \in \Gamma_L \cup \Gamma_R$ and arbitrary x_1 . As $(y_2 - x_2)^2$ lies in $\{z : \Im z \geq 0\} \setminus (-\infty, 0]$, it is easy to see that

$$0 \leq \Im \sqrt{x_1^2 + (y_2 - x_2)^2} \leq |\Im y_2|. \quad (6.3)$$

If $(x_1, x_2) \in K$, for a compact set $K \subset \mathbb{R}^2$, then as $|\Im y_2| \rightarrow \infty$, we see that

$$\frac{\Im \sqrt{x_1^2 + (y_2 - x_2)^2}}{|\Im y_2|} \rightarrow 1, \quad (6.4)$$

but note that as $|x_1| \rightarrow \infty$, with y_2 bounded, the $\Im \sqrt{x_1^2 + (y_2 - x_2)^2} \rightarrow 0^+$. In light of this we now assume that $\beta > 0$ in order to be certain that various integrals we consider are *uniformly*, absolutely convergent.

From the standard asymptotic expansion for the free-space kernel we see that for $(x_1, x_2) \in \mathbb{R}$, with $|x_2| < L_C$ and $y_2 \in \Gamma_L \cup \Gamma_R$, we obtain the uniform estimates

$$\left| G_0(x_1, x_2; 0; y_2) - \frac{C_{\pm} e^{ik\sqrt{x_1^2 + (x_2 - y_2)^2}}}{[x_1^2 + (x_2 - y_2)^2]^{\frac{1}{4}}} \right| \leq \frac{K e^{-k\Im \sqrt{x_1^2 + (x_2 - y_2)^2}}}{\sqrt{|x_1^2 + (x_2 - y_2)^2|}}; \quad (6.5)$$

with a similar estimate for the ∂_{y_1} -derivatives of the free-space kernel:

$$\left| \partial_{y_1} G_0(x_1, x_2; 0; y_2) - \frac{C'_{\pm} x_1 e^{ik\sqrt{x_1^2 + (x_2 - y_2)^2}}}{[x_1^2 + (x_2 - y_2)^2]^{\frac{3}{4}}} \right| \leq \frac{K(|x_1| + 1) e^{-k\Im \sqrt{x_1^2 + (x_2 - y_2)^2}}}{\sqrt{|x_1^2 + (x_2 - y_2)^2|^{\frac{5}{4}}}}. \quad (6.6)$$

Using these estimates we can deform the contour of integration for the free-space terms.

Lemma 12. *Let $\sigma \in \mathcal{C}^{\alpha,\beta}$ and $\tau \in \mathcal{C}^{\alpha+\frac{1}{2},\beta}$ for $0 < \alpha < \frac{1}{2}$, $0 \leq \beta \leq k$. The values $\mathcal{S}_{\tilde{\Gamma}}^{l,r}[\tau](\mathbf{x})$ and $\mathcal{D}_{\tilde{\Gamma}}^{l,r}[\sigma](\mathbf{x})$ exist for every $\mathbf{x} \in \mathbb{R} \times [-L_C, L_C]$ and are independent of $\tilde{\Gamma} \in \mathcal{G}$.*

Proof. We use the splitting

$$G^{l,r} = G_0 + w^{l,r} \quad (6.7)$$

to split the layer potentials into a free-space part and a waveguide correction:

$$\mathcal{S}_{\tilde{\Gamma}}^{l,r} = \mathcal{S}_{\tilde{\Gamma}} + \mathcal{S}_{w,\tilde{\Gamma}}^{l,r} \quad \text{and} \quad \mathcal{D}_{\tilde{\Gamma}}^{l,r} = \mathcal{D}_{\tilde{\Gamma}} + \mathcal{D}_{w,\tilde{\Gamma}}^{l,r}. \quad (6.8)$$

The existence and path independence of the waveguide contributions $\mathcal{S}_w^{l,r}$ and $\mathcal{D}_w^{l,r}$ is proved above. The proof of convergence and path independence of the free-space terms follows from (6.5), (6.3), the analyticity properties of $H_0^{(1)}(z)$, and the decay properties of (σ, τ) $\square$

This lemma combined with Theorem 1, Lemma 11, and Theorem 3 in [23] completes the proof of Theorem 2. We conclude this section by showing that various physically relevant incoming fields give data that satisfies the conditions of Theorem 2.

6.1 Admissible Data

We consider various types of physically relevant data for our integral equations. Some of the data is shown to be admissible. For certain cases the data is not obviously admissible, but it is explained how the data can be modified to make it admissible.

Point sources

If $\mathbf{y}$ is fixed with $y_1 \neq 0$, and $|y_2| < L_C$, then, as x_2 tends to infinity within $\Gamma_L \cup \Gamma_R$, we have the estimate

$$\sqrt{y_1^2 + (x_2 - y_2)^2} \sim (\text{sign } \Re x_2) x_2. \quad (6.9)$$

This observation, along with standard properties of Hankel functions show that the waveguide Green's function gives data in the desired $\mathcal{C}^{\alpha,\beta}$ spaces. The outgoing asymptotics for $w^{l,r}$ are derived in [20].

Lemma 13. *If $\mathbf{y} \in (\mathbb{R} \setminus \{0\}) \times [-L_C, L_C]$, then, as a function of x_2 ,*

$$f_D(x_2) = G^{l,r}(0, x_2; \mathbf{y}) \in \mathcal{C}^{\frac{1}{2},k} \quad \text{and} \quad f_N(x_2) = \partial_{x_1} G^{l,r}(0, x_2; \mathbf{y}) \in \mathcal{C}^{\frac{3}{2},k}. \quad (6.10)$$

Further, these f_D and f_N have outgoing asymptotic expansions like those in (3.13).

Waveguide modes

For any non-negative potential q , which is not identically zero, the equation $(\Delta + k^2(q+1))u = 0$ admits solutions of the form

$$u(x_1, x_2) = v(x_2)e^{\pm i\xi x_1}, \quad \text{with } v \in L^2(\mathbb{R}), \quad (6.11)$$

for a finite collections of $\xi \in (k, k\sqrt{M_q+1})$, where $M_q = \sup\{q(x)\}$. These solutions are known as waveguide modes for the potential q . Let $v(x_2)e^{\pm i\xi x_1}$ be a waveguide mode of $q = q^l$ or $q = q^r$. The corresponding boundary data for the integral equation is given by $f_D(x_2) = v(x_2)$ and $f_N(x_2) = \pm i\xi v(x_2)$. There are constants $C_{\pm}$ so that $v \in \mathcal{C}^1(\mathbb{R})$ is of the form

$$v(x_2) = C_{\pm} e^{\mp \sqrt{\xi^2 - k^2} x_2} \quad (6.12)$$

when $\pm x_2 > d$, (see [20]). This data is clearly analytic on the interior of Γ_C and decays exponentially as $|\Re x_2| \rightarrow \infty$, but not as $|\Im x_2| \rightarrow \infty$, so does not belong to any $\mathcal{C}^{\alpha,\beta}$ -space. This will prevent us

from applying the truncation estimate in Theorem 3. To overcome this defect we let $\tilde{f}_D$, and $\tilde{f}_N$ be given by

$$\begin{bmatrix} \tilde{f}_D \\ \tilde{f}_N \end{bmatrix} = -\mathcal{K}_{\mathbb{R}} \begin{bmatrix} f_D \\ f_N \end{bmatrix}. \quad (6.13)$$

Since $[f_D, f_N] \in \mathcal{C}_{\tilde{F}}^{\alpha, \beta} \oplus \mathcal{C}_{\tilde{F}}^{\alpha+1/2, \beta}$ for any $\tilde{F}$ which satisfy the slope condition in (3.12) and any $\alpha > 0$ and $\beta < \sqrt{\xi^2 - k^2}C$, it follows from Remark 7 that $[\tilde{f}_D, \tilde{f}_N] \in \mathcal{C}^{\alpha, k} \oplus \mathcal{C}^{\alpha+1/2, k}$, for any $\alpha \leq 1/2$. Let $[\tilde{\sigma}, \tilde{\tau}]$ denote the solution to (3.8) with data $[\tilde{f}_D, \tilde{f}_N]$, then

$$\begin{bmatrix} \sigma \\ \tau \end{bmatrix} = \begin{bmatrix} \tilde{\sigma} \\ \tilde{\tau} \end{bmatrix} + \begin{bmatrix} f_D \\ f_N \end{bmatrix} \quad (6.14)$$

satisfies (3.8) for all $\tilde{F} \in \mathcal{G}$ which satisfy the slope condition in (3.12). Further, it is also clear that

$$\mathcal{S}_{\tilde{F}}^{l,r}[f_N](\mathbf{x}) + \mathcal{D}_{\tilde{F}}^{l,r}[f_D](\mathbf{x}) = \mathcal{S}_{\mathbb{R}}^{l,r}[f_N](\mathbf{x}) + \mathcal{D}_{\mathbb{R}}^{l,r}[f_D](\mathbf{x}), \quad (6.15)$$

so Theorem 2 can be extended to this case.

6.2 Truncation error

In this section, we show that we can solve (3.8) on a truncated contour with a controllable error. We fix a $\tilde{F} \in \mathcal{G}$. Assuming that $\beta > 0$, we let $\tilde{F}_\epsilon$ be the truncation of $\tilde{F}$ to the region $\exp(-\beta|\Im x_2|) < \epsilon$. We can think of $\mathcal{K}_{\tilde{F}}, \mathcal{K}_{\tilde{F}_\epsilon}$ either as maps

$$\mathcal{K}_{\tilde{F}}, \mathcal{K}_{\tilde{F}_\epsilon} : \mathcal{C}_{\tilde{F}}^{\alpha, \beta} \oplus \mathcal{C}_{\tilde{F}}^{\alpha+1/2, \beta} \rightarrow \mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha+1/2, \beta}, \quad (6.16)$$

or, after first composing with restriction to $\tilde{F}$ or $\tilde{F}_\epsilon$, respectively, as maps

$$\mathcal{K}_{\tilde{F}}, \mathcal{K}_{\tilde{F}_\epsilon} : \mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha+1/2, \beta} \rightarrow \mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha+1/2, \beta}. \quad (6.17)$$

For the most part, the latter interpretation is more useful in our applications.

We begin our analysis of truncation error by showing that, for sufficiently small $0 < \epsilon$, the difference between $\mathcal{K}_{\tilde{F}}$ and $\mathcal{K}_{\tilde{F}_\epsilon}$ can be bounded in norm by ϵ with a prefactor depending on the potentials, $q_{l,r}$, and the choice of contour $\tilde{F}$.

Lemma 14. *For $0 < \alpha < \frac{1}{2}$, $-k < \beta \leq k$, $0 < \epsilon$, and $\tilde{F} \in \mathcal{G}$, let*

$$\tilde{F}_\epsilon = \{x_2 \in \tilde{F} : e^{-(\beta+k)|\Im x_2|} < \epsilon\}.$$

There exists a $C > 0$ such that

$$\left\| (\mathcal{K}_{\tilde{F}} - \mathcal{K}_{\tilde{F}_\epsilon}) \begin{bmatrix} \sigma_{\tilde{F}} \\ \tau_{\tilde{F}} \end{bmatrix} \right\|_{\mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha+1/2, \beta}} < C\epsilon \left\| \begin{bmatrix} \sigma_{\tilde{F}} \\ \tau_{\tilde{F}} \end{bmatrix} \right\|_{\mathcal{C}_{\tilde{F}}^{\alpha, \beta} \oplus \mathcal{C}_{\tilde{F}}^{\alpha+1/2, \beta}}. \quad (6.18)$$

Proof. We begin by looking at the operator C . Letting σ be any element of $\mathcal{C}^{\alpha, \beta}$, we have

$$C_{\tilde{F}}[\sigma](x_2) - C_{\tilde{F}_\epsilon}[\sigma](x_2) = \int_{\tilde{F} \setminus \tilde{F}_\epsilon} k_C(x_2, y_2) \sigma(y_2) dy_2. \quad (6.19)$$

Using our bounds on k_C and σ gives

$$\begin{aligned} \left| C_{\tilde{F}}[\sigma](x_2) - C_{\tilde{F}_\epsilon}[\sigma](x_2) \right| &\leq \int_{\tilde{F} \setminus \tilde{F}_\epsilon} \frac{K e^{-k(|\Im x_2| + |\Im y_2|)} e^{-\beta|\Im y_2|} \|\sigma\|_{\alpha, \beta, \tilde{F}}}{(1 + |x_2| + |y_2|)^{\frac{3}{2}} (1 + |y_2|)^\alpha} |dy_2| \\ &\leq \int_{\tilde{F} \setminus \tilde{F}_\epsilon} \frac{\epsilon K e^{-k|\Im x_2|} \|\sigma\|_{\alpha, \beta, \tilde{F}}}{(1 + |x_2| + |y_2|)^{\frac{3}{2}} (1 + |y_2|)^\alpha} |dy_2|. \end{aligned} \quad (6.20)$$

We let $s(y_2)$ be the signed arc-length from 0 to y_2 along $\tilde{\Gamma}$; the monotonicity condition in (3.1) implies that that $|y_2| > \frac{1}{\sqrt{2}}|s|$. This allows us to bound the integral by

$$\left| C_{\tilde{\Gamma}}[\sigma](x_2) - C_{\tilde{\Gamma}_\epsilon}[\sigma](x_2) \right| \leq \int_{-\infty}^{\infty} \frac{\epsilon \sqrt{2} K e^{-k|\Im x_2|} \|\sigma\|_{\alpha, \beta, \tilde{\Gamma}}}{(1 + |x_2| + \frac{|s|}{\sqrt{2}})^{\frac{3}{2}} (1 + \frac{|s|}{\sqrt{2}})^\alpha} ds \quad (6.21)$$

Applying Lemma 2 gives the estimate

$$\|C_{\tilde{\Gamma}}[\sigma] - C_{\tilde{\Gamma}_\epsilon}[\sigma]\|_{\alpha + \frac{1}{2}, \beta} < C\epsilon \|\sigma\|_{\alpha, \beta, \tilde{\Gamma}}.$$

A similar argument can be applied to D to show that

$$\|D_{\tilde{\Gamma}}[\tau] - D_{\tilde{\Gamma}_\epsilon}[\tau]\|_{\alpha, \beta} < C\epsilon \|\tau\|_{\alpha + \frac{1}{2}, \beta, \tilde{\Gamma}}. \quad (6.22)$$

Combining these estimates gives the desired result. $\square$

For $(\sigma, \tau) \in \mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}$ we therefore have the estimate

$$\left\| (\mathcal{K}_{\tilde{\Gamma}} - \mathcal{K}_{\tilde{\Gamma}_\epsilon}) \begin{bmatrix} \sigma_{\tilde{\Gamma}} \\ \tau_{\tilde{\Gamma}} \end{bmatrix} \right\|_{\mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}} < C\epsilon \left\| \begin{bmatrix} \sigma \\ \tau \end{bmatrix} \right\|_{\mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}}. \quad (6.23)$$

Using this estimate, we can control the error caused by solving (3.8) on a truncated contour.

Theorem 7. *For each $0 < \alpha < \frac{1}{2}$, $-k < \beta \leq k$, and $\tilde{\Gamma} \in \mathcal{G}$, there is an $\epsilon_0 > 0$, such that, for $\epsilon < \epsilon_0$, the operator*

$$(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}_\epsilon}) : \mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta} \rightarrow \mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta} \quad (6.24)$$

is invertible and there exists a $C > 0$, independent of ϵ , such that

$$\left\| \left[(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}})^{-1} - (\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}_\epsilon})^{-1} \right] \begin{bmatrix} f_D \\ f_N \end{bmatrix} \right\|_{\mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}} < C\epsilon \left\| \begin{bmatrix} f_D \\ f_N \end{bmatrix} \right\|_{\mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}}. \quad (6.25)$$

Proof. We first note that

$$\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}_\epsilon} = \mathcal{I} + \mathcal{K}_{\tilde{\Gamma}} + (\mathcal{K}_{\tilde{\Gamma}_\epsilon} - \mathcal{K}_{\tilde{\Gamma}}) = (\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}}) \left[\mathcal{I} + (\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}})^{-1} (\mathcal{K}_{\tilde{\Gamma}_\epsilon} - \mathcal{K}_{\tilde{\Gamma}}) \right], \quad (6.26)$$

where we have used the fact that $\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}}$ is invertible on $\mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}$, which contains the range of $\mathcal{K}_{\tilde{\Gamma}_\epsilon} - \mathcal{K}_{\tilde{\Gamma}}$. The estimate in (6.23) shows that there is a constant, C , so that the norm of the composition

$$(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}})^{-1} (\mathcal{K}_{\tilde{\Gamma}_\epsilon} - \mathcal{K}_{\tilde{\Gamma}}) : \mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta} \rightarrow \mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta} \quad (6.27)$$

is bounded by

$$\eta = C\epsilon \left\| (\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}})^{-1} \right\|_{\mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}}.$$

For sufficiently small $0 < \epsilon$, we see that $\eta < 1$; we can therefore use a Neumann series to invert the term in brackets on the right hand side of (6.26) to obtain the invertibility of $\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}_\epsilon}$, as well as the expression

$$(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}_\epsilon})^{-1} = \left[\mathcal{I} + (\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}})^{-1} (\mathcal{K}_{\tilde{\Gamma}_\epsilon} - \mathcal{K}_{\tilde{\Gamma}}) \right]^{-1} (\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}})^{-1}. \quad (6.28)$$

From this formula and the Neumann series we easily show the existence of a constant C' for which

$$\left\| \left[(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}_\epsilon})^{-1} - (\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}})^{-1} \right] \begin{bmatrix} f_D \\ f_N \end{bmatrix} \right\|_{\mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}} \leq C'\epsilon \left\| \begin{bmatrix} f_D \\ f_N \end{bmatrix} \right\|_{\mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}}, \quad (6.29)$$

which is equivalent to the desired result. $\square$

We now observe that it is enough to enforce the integral equation on the compact contour $\tilde{\Gamma}_\epsilon$.

Lemma 15. *Let α, β , and ϵ_0 satisfy the assumptions of the previous theorem. If $\tilde{\Gamma} \in \mathcal{G}$ and $[f_D, f_N] \in \mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}$, then for any $\epsilon < \epsilon_0$ there exists a unique $[\sigma_\epsilon, \tau_\epsilon] \in \mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}$ satisfying*

$$\left(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}_\epsilon} \right) \begin{bmatrix} \sigma_\epsilon \\ \tau_\epsilon \end{bmatrix} (x_2) = \begin{bmatrix} f_D \\ f_N \end{bmatrix} (x_2) \quad \forall x_2 \in \tilde{\Gamma}_\epsilon. \quad (6.30)$$

Moreover, the solution to (6.30) agrees with

$$\left(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}_\epsilon} \right)^{-1} \begin{bmatrix} f_D \\ f_N \end{bmatrix}, \quad (6.31)$$

with $\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}_\epsilon}$ being the operator defined on the spaces in (6.24).

Proof. The existence of a solution is an immediate consequence of the previous theorem, i.e., a solution obtained by enforcing the integral equation for all $x_2 \in \Gamma_\mathbb{C}$ trivially satisfies the equation on $\tilde{\Gamma}_\epsilon$. Now let $[\sigma_\epsilon, \tau_\epsilon]$ denote any solution obtained by enforcing the integral equation for all $x_2 \in \tilde{\Gamma}_\epsilon$. To prove uniqueness, we let

$$\begin{bmatrix} \tilde{\sigma}_\epsilon \\ \tilde{\tau}_\epsilon \end{bmatrix} := \begin{bmatrix} f_D \\ f_N \end{bmatrix} - \mathcal{K}_{\tilde{\Gamma}_\epsilon} \begin{bmatrix} \sigma_\epsilon \\ \tau_\epsilon \end{bmatrix}, \quad (6.32)$$

for all $x_2 \in \Gamma_\mathbb{C}$. This function is in $\mathcal{C}^{\alpha, \beta} \oplus \mathcal{C}^{\alpha + \frac{1}{2}, \beta}$ by Lemma 6 and is equal to $[\sigma_\epsilon, \tau_\epsilon]$ on $\tilde{\Gamma}_\epsilon$ by (6.30). Moreover, it satisfies

$$\left(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}_\epsilon} \right) \begin{bmatrix} \tilde{\sigma}_\epsilon \\ \tilde{\tau}_\epsilon \end{bmatrix} = \begin{bmatrix} f_D \\ f_N \end{bmatrix} \quad (6.33)$$

on $\Gamma_\mathbb{C}$. Theorem 7 gives that $\tilde{\sigma}_\epsilon$ and $\tilde{\tau}_\epsilon$ are both uniquely determined by f_D and f_N . The uniqueness of $[\sigma_\epsilon, \tau_\epsilon]$ therefore follows from the identity theorem for analytic functions. $\square$

We now give a proof of Theorem 3, which bounds the error in the potential u_ϵ obtained by solving for densities $(\sigma_\epsilon, \tau_\epsilon)$ on any truncated curve $\tilde{\Gamma}_\epsilon$. Since we now include the contribution of the free-space kernel, we again assume that $0 < \beta$.

Proof of Theorem 3. Recall that $0 < \beta$ and $\tilde{\Gamma}_\epsilon^c = \{y_2 \in \tilde{\Gamma} : e^{-\beta|\Im y_2|} < \epsilon\}$. Let $[\sigma_\epsilon, \tau_\epsilon]$ solve (6.30) and $[\sigma, \tau]$ solve the equation

$$\left(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}} \right) \begin{bmatrix} \sigma \\ \tau \end{bmatrix} = \begin{bmatrix} f_D \\ f_N \end{bmatrix} \quad (6.34)$$

on all of $\Gamma_\mathbb{C}$. By Theorem 7 and Lemma 15, we have that

$$\|\sigma - \sigma_\epsilon\|_{\alpha, \beta} + \|\tau - \tau_\epsilon\|_{\alpha + \frac{1}{2}, \beta} < C\epsilon \left(\|f_D\|_{\alpha, \beta} + \|f_N\|_{\alpha + \frac{1}{2}, \beta} \right). \quad (6.35)$$

We then write

$$\mathcal{S}_{\tilde{\Gamma}_\epsilon}^l[\tau_\epsilon] - \mathcal{S}_{\tilde{\Gamma}}^l[\tau] = \mathcal{S}_{\tilde{\Gamma}}^l[\tau_\epsilon - \tau] + \left(\mathcal{S}_{\tilde{\Gamma}_\epsilon}^l - \mathcal{S}_{\tilde{\Gamma}}^l \right) [\tau_\epsilon]. \quad (6.36)$$

Using Lemma 22 and (6.5) we can prove that

$$\begin{aligned} \left| \left(\mathcal{S}_{\tilde{\Gamma}_\epsilon}^l - \mathcal{S}_{\tilde{\Gamma}}^l \right) [\tau_\epsilon](\mathbf{x}) \right| &\leq C\epsilon \|\tau_\epsilon\|_{\alpha + \frac{1}{2}, \beta} \\ &\leq C'\epsilon \left(\|f_D\|_{\alpha, \beta} + \|f_N\|_{\alpha + \frac{1}{2}, \beta} \right), \end{aligned} \quad (6.37)$$

with a similar proof to Lemma 14. Using the same steps as the proof of Lemma 6 and Theorem 7, we can show that

$$\begin{aligned} |\mathcal{S}_{\tilde{\Gamma}}^l[\tau_\epsilon - \tau](\mathbf{x})| &\leq C\|\tau_\epsilon - \tau\|_{\alpha + \frac{1}{2}, \beta} \\ &\leq C'\epsilon \left(\|f_D\|_{\alpha, \beta} + \|f_N\|_{\alpha + \frac{1}{2}, \beta} \right). \end{aligned} \quad (6.38)$$

A similar proof shows that

$$|\mathcal{D}_{\tilde{\Gamma}_\epsilon}^l[\sigma_\epsilon](\mathbf{x}) - \mathcal{D}_{\tilde{\Gamma}}^l[\sigma](\mathbf{x})| < C'\epsilon \left(\|f_D\|_{\alpha,\beta} + \|f_N\|_{\alpha+\frac{1}{2},\beta} \right). \quad (6.39)$$

Summing these two bounds and applying Theorem 2 gives that

$$\begin{aligned} \left| \mathcal{S}_{\tilde{\Gamma}_\epsilon}^l[\tau_\epsilon] - \mathcal{D}_{\tilde{\Gamma}_\epsilon}^l[\sigma_\epsilon](\mathbf{x}) - u^l(\mathbf{x}) \right| &= \left| \mathcal{S}_{\tilde{\Gamma}_\epsilon}^l[\tau_\epsilon] - \mathcal{D}_{\tilde{\Gamma}_\epsilon}^l[\sigma_\epsilon](\mathbf{x}) - \mathcal{S}_{\tilde{\Gamma}}^l[\tau] + \mathcal{D}_{\tilde{\Gamma}}^l[\sigma](\mathbf{x}) \right| \\ &< 2C\epsilon \left(\|f_D\|_{\alpha,\beta} + \|f_N\|_{\alpha+\frac{1}{2},\beta} \right). \end{aligned} \quad (6.40)$$

Repeating the same argument for u^r gives the desired result. $\square$

We conclude this section with the proof of Corollary 1, which addresses the problems associated with waveguide mode data.

Proof of Corollary 1. Rearranging (3.16) shows that $[\tilde{\sigma}, \tilde{\tau}]$ must solve

$$\left(\mathcal{I} + \mathcal{K}_{\tilde{\Gamma}} \right) \begin{bmatrix} \tilde{\sigma} \\ \tilde{\tau} \end{bmatrix} (x_2) = -\mathcal{K}_{\tilde{\Gamma}} \begin{bmatrix} f_D \\ f_N \end{bmatrix} (x_2) \quad \forall x_2 \in \tilde{\Gamma}. \quad (6.41)$$

We observe that the right hand side is in $\mathcal{C}^{1/2,k} \oplus \mathcal{C}^{3/2,k}$. The existence and uniqueness of $[\tilde{\sigma}, \tilde{\tau}]$ then follows from Theorem 1. The formula (3.18) then follows from Lemma 12 and the analyticity of the waveguide modes outside the support of $q^{l,r}$. Theorem 3 in [23] then gives that the function $u^{l,r}$ solves (2.5).

The error estimate (3.20) cannot be derived from Theorem 3 because though $[f_D, f_N]$ are in $\mathcal{C}^{0,0} \oplus \mathcal{C}^{0,0}$, but they are not in any $\mathcal{C}^{\alpha,0}$ for $\alpha > 0$. Instead, we apply Theorem 3 with $\beta = k$ to see that

$$\begin{aligned} \left| \left(\mathcal{S}_{\tilde{\Gamma}_\eta}^{l,r}[\tilde{\tau}_\epsilon](\mathbf{x}) + \mathcal{D}_{\tilde{\Gamma}_\eta}^{l,r}[\tilde{\sigma}_\epsilon](\mathbf{x}) \right) - \left(\mathcal{S}_{\tilde{\Gamma}}^{l,r}[\tilde{\tau}](\mathbf{x}) + \mathcal{D}_{\tilde{\Gamma}}^{l,r}[\tilde{\sigma}](\mathbf{x}) \right) \right| &< C\eta \left\| \mathcal{K}_{\tilde{\Gamma}} \begin{bmatrix} f_D \\ f_N \end{bmatrix} \right\|_{\mathcal{C}^{1/2,k} \oplus \mathcal{C}^{1,k}} \\ &\leq C\epsilon \left\| \mathcal{K}_{\tilde{\Gamma}} \begin{bmatrix} f_D \\ f_N \end{bmatrix} \right\|_{\mathcal{C}^{1/2,k} \oplus \mathcal{C}^{1,k}}. \end{aligned} \quad (6.42)$$

What remains is to bound the truncation error for $\mathcal{S}_{\tilde{\Gamma}_\eta}^{l,r}[f_N]$ and $\mathcal{D}_{\tilde{\Gamma}_\eta}^{l,r}[f_D]$. For any fixed $\tilde{\Gamma}$ satisfying the slope condition (3.12), it is easy to show that the data $[f_D, f_N]$ restricted to $\tilde{\Gamma}$ is exponentially decaying, and therefore lies in $\mathcal{C}_{\tilde{\Gamma}}^{\alpha,\beta} \oplus \mathcal{C}_{\tilde{\Gamma}}^{\alpha+\frac{1}{2},\beta}$ for some positive α and β . The value of β depends on the slope of the curve $\tilde{\Gamma}$ and the waveguide frequency ξ . The estimate for the single layer contribution is

$$\begin{aligned} \left| \mathcal{S}_{\tilde{\Gamma}_\eta}^{l,r}[f_N](\mathbf{x}) - \mathcal{S}_{\tilde{\Gamma}}^{l,r}[f_N](\mathbf{x}) \right| &\leq \int_{\tilde{\Gamma} \setminus \tilde{\Gamma}_\eta} |G^{l,r}(\mathbf{x}; 0; y_2) f_N(y_2)| |dy_2| \\ &\leq C \int_{\tilde{\Gamma} \setminus \tilde{\Gamma}_\eta} e^{-\beta|\Im y_2|} |dy_2|. \end{aligned} \quad (6.43)$$

Let η_0 be such that the above integral is less than ϵ . The parameter η is then given by $\eta = \min(\eta_0, \epsilon)$. This issue only arises if x_1 is allowed to tend to infinity. It has nothing to do with either the solvability of the integral equation, or the truncation error in (6.42). A similar argument applies to the double layer contribution. $\square$

7 Numerical experiments

In this section, we illustrate the performance of our approach through several numerical examples. We set $k = 3$ throughout and consider two different configurations: a) where $q_{l,r}^a(x_2)$ are the smooth potentials given by

$$\begin{aligned} q_l^a(x) &= 2e^{-2x^2}(3 + 2\cos(5\pi x + 0.7) - \cos(2x + 1.6)), \\ q_r^a(x) &= \frac{8}{3}e^{-x^2/0.245}(1.53 - \cos(1.05) + 0.53\cos(x + 0.25)), \end{aligned} \quad (7.1)$$

and; b) where $q_l^b = q_l^a$ and $q_r^b = \chi_{[-3,3]}$. The parameters are chosen so that $q_l := q_l^{a,b}$ supports four waveguide modes with frequencies

$$(\xi_{l,1}, \xi_{l,2}, \xi_{l,3}, \xi_{l,4}) = (3.7401, 4.6783, 6.2958, 7.5942),$$

and $q_r^{a,b}$ support two, respectively five waveguide modes with frequencies

$$\begin{aligned} (\xi_{r,1}^a, \xi_{r,2}^a) &= (4.1378, 5.9297), \\ (\xi_{r,1}^b, \xi_{r,2}^b, \xi_{r,3}^b, \xi_{r,4}^b, \xi_{r,5}^b) &= (3.2274, 3.5503, 3.8083, 4.0022, 4.2164). \end{aligned}$$

For brevity, we have reported the first 5 significant digits of the waveguide frequencies, however, they have been computed to 10 significant digits in the numerical computations. The potentials and waveguide modes are plotted in Figure 3.

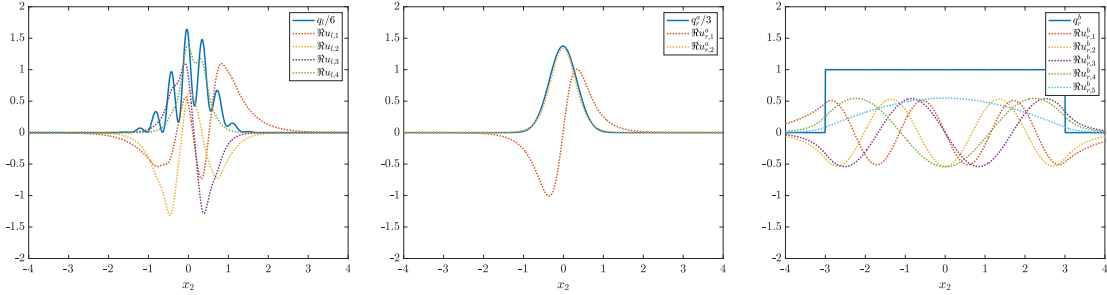


Figure 3: The potentials q_l and $q_r^{a,b}$, shown in blue, and cross-sections of the corresponding waveguide modes.

The efficient evaluation of the Green's functions $G^{l,r}$ is fairly involved. It is perhaps most natural to compute them via their partial Fourier transform. There are three main technical issues: the slow decay of the Fourier transform induced by the logarithmic singularity in real space for $x_2, y_2 \in [-d, d]$; the presence of waveguide modes, which correspond to poles on the real axis in the Fourier domain, and the oscillatory nature of the integrand for large $|x_1 - y_1|$. Here we provide a brief sketch of the method used in this paper for their computation. We refer the reader to [8–10, 24, 40, 42, 51, 56], for examples of other approaches.

As mentioned above, the Green's functions $G^{l,r}$ are evaluated by taking the Fourier transform in the x_1 variable. The Fourier transform is then evaluated at interpolation nodes on a complex contour, similar to the blue contour in Figure 9. For each Fourier frequency ξ , depending on y_2 , the solution for all x_2 and $x_1 - y_1$ requires the solution of an ordinary differential equation (ODE) on $\mathbb{R}$ (or more precisely $\tilde{\mathbb{R}}$, though this does not require extra work). This ODE is solved using a shooting method with a Runge-Kutta discretization when $|\xi|$ is small, and using a WKB expansion of order 6 when $|\xi|$ is large when q is smooth, and a WKB expansion of order 1 for the piecewise constant case. The discretization of the contour and the parameters of the ODE solver are chosen to guarantee 6 relative digits of accuracy in the computed Green's function for $\mathbf{x}, \mathbf{y} \in [-L_0, L_0] \times \Gamma_{\mathbb{C}}$.

In Figure 4, we plot the Green's functions for the three different media for $\mathbf{y}_0 = (0.1, 1)$. The accuracy of the Green's function evaluator is measured in two ways. In Figure 5 (left panel), we plot

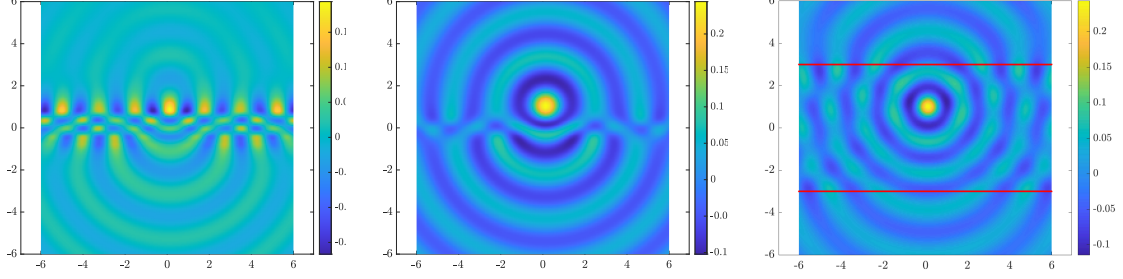


Figure 4: Imaginary part of the Green's functions $G_q(\mathbf{x}, \mathbf{y}_0)$ with $q = q_l, q_r^a, q_r^b$ (left to right), and $\mathbf{y}_0 := (0.1, 1)$.

the residual $|(\Delta + k^2(1+q))G|$ which should be 0 away from the point source $\mathbf{y}_0$. We also verify the accuracy of the Green's function by using Green's identities. Suppose $u(\mathbf{x}) = G_q(\mathbf{x}, \mathbf{y}_0)$, where G_q is the Green's function with material property q , then u for $|x_2| < L_C$ satisfies

$$\int_{\tilde{\Gamma}_\epsilon} G_q(\mathbf{x}; 0, y_2) \frac{\partial u}{\partial y_1}(0, y_2) dy_2 - \int_{-\tilde{\Gamma}_\epsilon} \frac{\partial G_q}{\partial y_1}(\mathbf{x}; 0, y_2) u(0, y_2) dy_2 = \begin{cases} u(\mathbf{x}) & x_1 < 0, \\ u(\mathbf{x})/2 & x_1 = 0, \\ 0 & x_1 > 0. \end{cases} \quad (7.2)$$

Here $\tilde{\Gamma} \in \mathcal{G}$ is the complex contour given by

$$\tilde{\Gamma} = t + i\psi(t), \quad (7.3)$$

with

$$\psi(t) = 20 \left(\text{Erf} \left(\frac{40+t}{5} \right) - \text{Erf} \left(\frac{40-t}{5} \right) \right), \quad (7.4)$$

with the curve $\tilde{\Gamma}_\epsilon$ truncated at $|\Im(x_2)| = 39.527$. The error in Figure 5 (right panel) is the absolute deviation in the validity of the above identity.

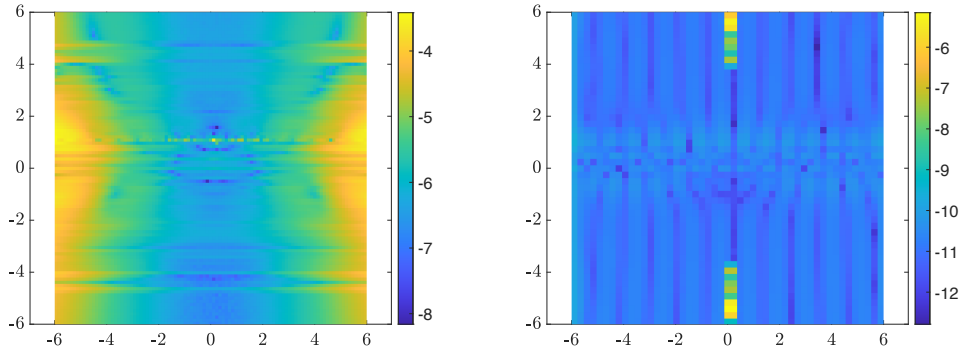


Figure 5: Left: $\log_{10}$ of the residual $(\Delta_h + k^2(1 + q_l(x_2)))G_{q_l}(\mathbf{x}, \mathbf{y}_0)$ associated with the computed Green's function for the left material, with $\mathbf{y}_0 = (0.1, 1)$, and Δ_h the standard finite-difference approximation to the Laplacian with step-size $h = 10^{-4}$. Right: $\log_{10}$ of the computed difference between the left- and right-hand sides of (7.2) for $q = q_l$.

The layer potentials above and the ones appearing in the numerical solution of integral equations in the remainder of this section are discretized using the chunkIE package [2]. This package solves integral equations using a modified Nyström method. It discretizes $\tilde{\Gamma}_\epsilon$ by splitting it into 16th order Gauss-Legendre panels and provides utilities for the accurate evaluation of integral operators with logarithmic singularities on surface. The smooth quadrature rule is used to evaluate the integrals off-surface.

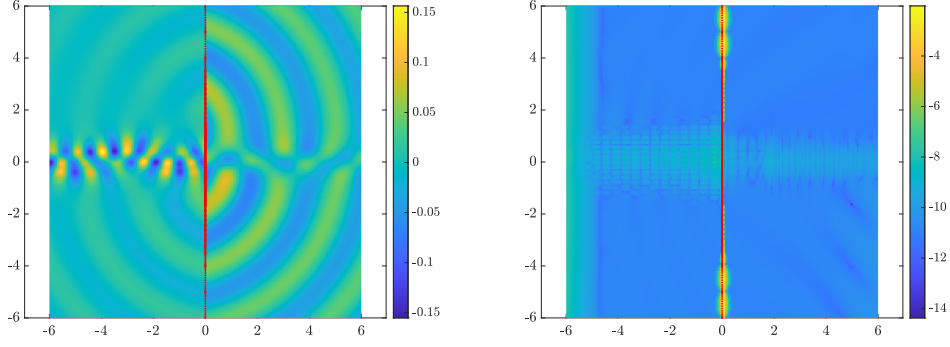


Figure 6: The left panel shows the incoming field, with $u_i^{l,r}$ respectively corresponding to point sources at $(1, 0.5)$ and $(-1.5, 1)$ for the analytic solution test. The absolute error of the solution measured by $\log_{10}(|u|)$ is plotted in the right panel.

Next we turn our attention to transmission problem. Throughout, we represent our solution u to the PDE (1.1) as

$$u(\mathbf{x}) = u_i^{l,r}(\mathbf{x}) + u^{l,r}(\mathbf{x}), \quad \text{for } \mathbf{x} \in \Omega_{l,r},$$

where $u_i^{l,r}$ are the prescribed incoming fields, and the unknown functions $u^{l,r}$ satisfy (2.5). We then use the above Green's function evaluators to discretize (3.15), which allows us to obtain the densities σ_ϵ and τ_ϵ . The functions $u^{l,r}$ are evaluated by discretizing

$$u^{l,r}(\mathbf{x}) = \mathcal{S}_{\Gamma_\epsilon}^{l,r}[\sigma_\epsilon](\mathbf{x}) - \mathcal{D}_{\Gamma_\epsilon}^{l,r}[\tau_\epsilon](\mathbf{x})$$

using the smooth Gauss-Legendre quadrature rule for $x_2 \neq 0$.

To test the accuracy, we set

$$u_i^{l,r}(\mathbf{x}) = \int_{\mathbb{R}^2} G^{l,r}(\mathbf{x}, \mathbf{y}) f^{l,r}(\mathbf{y}) d\mathbf{y}, \quad \text{for } \mathbf{x} \in \Omega_{l,r},$$

where $f^{l,r}(\mathbf{y}) := \delta(\mathbf{y} - \mathbf{y}^{l,r})$, with $\mathbf{y}^l \in \Omega_r$ and $\mathbf{y}^r \in \Omega_l$. These fictitious point sources produce an outgoing solution $u \in \mathcal{C}^1(\mathbb{R}^2)$ to $\Delta_x u^{l,r}(\mathbf{x}) + k^2(1 + q_{l,r}(x_2))u^{l,r}(\mathbf{x}) = 0$ in $\Omega_{l,r}$, meaning that $u \equiv 0$. The right panel of Figure 6 illustrates the error via the computed absolute value of u when $\mathbf{y}^l = (1, 0.5)$ and $\mathbf{y}^r = (-1.5, 1)$. The corresponding incoming fields $u_i^{l,r}$ are plotted in the left panel.

Having tested our solver, we now use it to simulate scattering with a few choices of physically relevant data. In Figure 7, we plot the solution to the PDE (1.1) due to a point source $f(\mathbf{x}) := \delta(\mathbf{x} - \mathbf{y}_0)$ located at $\mathbf{y}_0 = (-1.5, 1)$. In Figure 8, we plot the solution of the PDE corresponding to waveguide-mode incident fields,

$$(u_i^l(\mathbf{x}), u_i^r(\mathbf{x})) = (v_{l,j}(x_2)e^{i\xi_{l,j}x_1}, 0) \quad \text{or} \quad (u_i^l(\mathbf{x}), u_i^r(\mathbf{x})) = (0, v_{r,j}(x_2)e^{-i\xi_{r,j}^b x_1})$$

for various waveguide mode frequencies $\xi_{l,j}$ and $\xi_{r,j}^b$.

8 Concluding remarks

In this work, we analyze the complex scaling approach applied to boundary integral equations for the solution of time harmonic acoustic wave scattering from junctions of dielectric ‘leaky’ waveguides proposed in [20, 21, 23]. The method in these papers involved the solution of a certain integral equation with unknown densities supported on an infinite fictitious interface separating the two waveguides. This led to a system of integral equations which, while Fredholm on the interface, have

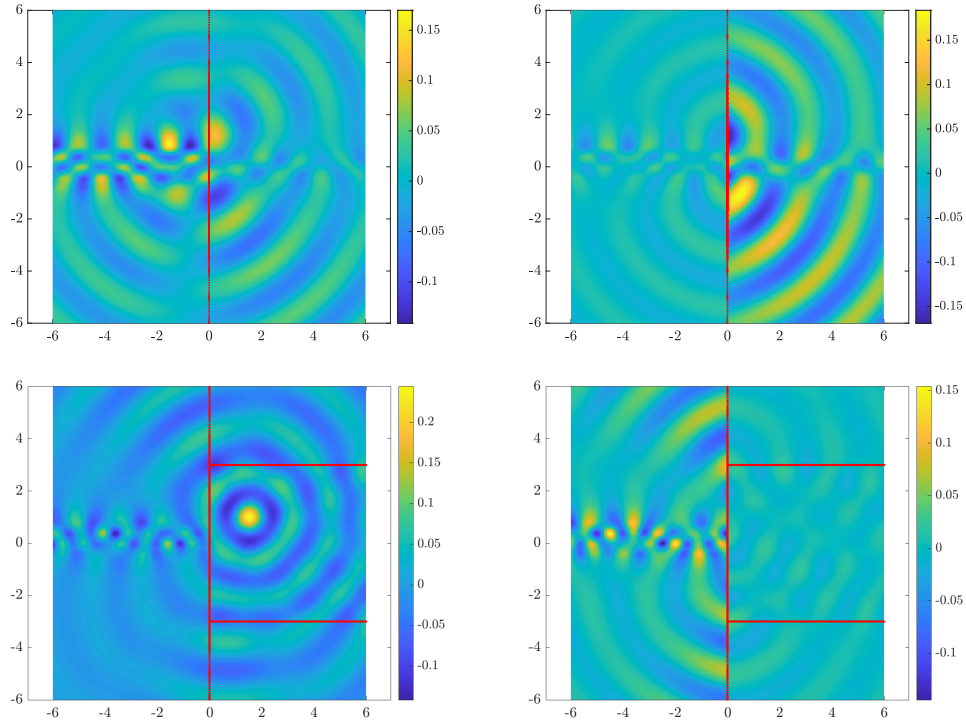


Figure 7: Top: Imaginary parts of the total and scattered fields corresponding to a point source at $\mathbf{y}_0 = (-1.5, 1)$, with potential $q_r = q_r^a$ in the right half-plane. Bottom: Imaginary parts of the total and scattered fields corresponding to a point source at $\mathbf{y}_0 = (1.5, 1)$, with potential $q_r = q_r^b$ in the right half-plane.

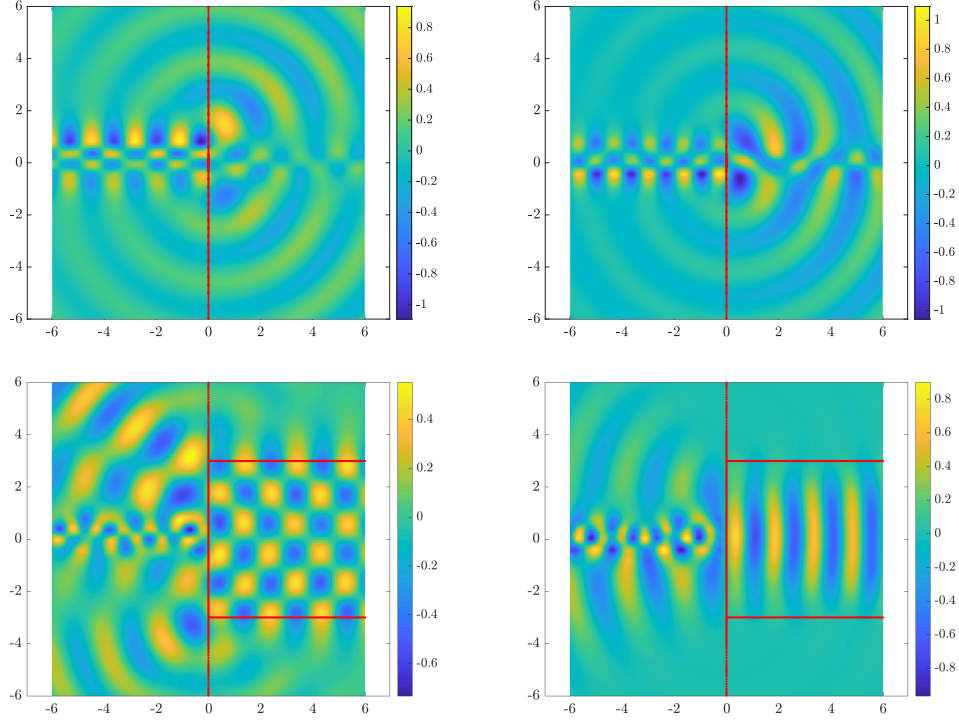


Figure 8: Imaginary parts of the solution corresponding to various incoming waveguide modes. Top: the potential in the right half-plane is $q_r = q_r^a$. The incoming waveguide mode is supported in the left half-plane, and has frequency $\xi_{l,1}, \xi_{l,2}$ in the left and right panels, respectively. Bottom: the potential in the right half-plane is $q_r = q_r^b$. The incoming waveguide mode is supported in the right half-plane, and has frequency $\xi_{r,1}^b, \xi_{r,5}^b$ in the left and right panels, respectively.

slowly decaying solutions, posing significant numerical challenges for their solution. In [24], it was shown through extensive numerical experiments that complex scaling is an effective approach for solving integral equations of this type. This work provides a rigorous justification for the use of the complex scaling approach in this context. In particular, we showed that the outgoing nature of both the Green's functions and the data can be used to analytically extend integral operators to appropriately chosen totally real submanifolds of $\mathbb{C}^2$ on which the data and the solutions to the integral equations decay rapidly. Moreover, we also showed that these contours lead to efficient discretizations of the integral equation and allow for truncation at $O(\log(1/\epsilon))$ wavelengths outside the support of the waveguides to obtain an ϵ accurate solution to the PDE.

There are many natural extensions of this work, both numerically and analytically. Firstly, in the numerical examples presented in this paper, the dominant computational cost was the efficient evaluation of Green's functions for bi-infinite waveguides. While there exist many methods for computing the Green's functions for the piecewise constant case [13], obtaining an efficient evaluator for generic perturbations q in the limits $x_2 \rightarrow y_2$, and $|x_1| \rightarrow \infty$ remains an open problem. Secondly, we believe that the analytical framework presented here can be extended to dielectric leaky waveguides for time-harmonic electromagnetic scattering, periodically varying waveguides and scattering from gratings, and interface problems arising in the study of topological insulators (see [3, 4] for an alternate approach). In principle, it should also be possible to extend it to three dimensions where quadrature [26] and fast algorithms are available. Finally, an interesting extension, which the current work does not directly address, would be the analysis of Fredholm properties of junctions that meet obliquely at the interface $\{x_1 = 0\}$; this would enable the analysis and the development of numerical methods for photonic device components such as 'Y-couplers'. Provided that all the wave guides are asymptotic to parallel rays, this more realistic problem is a relatively compact perturbation of the problems analyzed herein.

9 Acknowledgments

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A Properties of the partial Fourier transforms of $w^{l,r}$

Recall that the kernels $w^{l,r}$ are functions of $x_1 - y_1$ and thus their analytical properties can be understood by studying their partial Fourier transform $\tilde{w}^{l,r}$, given by

$$w^{l,r}(\mathbf{x}; \mathbf{y}) = \int_{\gamma} e^{i\xi(x_1 - y_1)} \tilde{w}^{l,r}(\xi; x_2, y_2) d\xi, \quad (\text{A.1})$$

for an appropriately chosen contour γ . In this section, we establish several key analytical properties of the functions $\tilde{w}^{l,r}$. For notational convenience, we drop the superscripts l, r . In particular, let q be a piecewise continuous function with compact support in $[-d, d]$, let $G = G_0 + w$ denote the outgoing Green's function of $\Delta + k^2(1 + q(x_2))$, where G_0 is, as before, the free-space Green's function for wave number k^2 . Finally let $\tilde{w}$ denote the partial Fourier transform of w in the x_1 -direction.

We denote the four open quadrants of the complex plane by Q_1, Q_2, Q_3, Q_4 , i.e., $Q_1 = \{z : \Re z > 0, \Im z > 0\}$ and $Q_j = e^{\frac{i(j-1)\pi}{2}} Q_1$, for $j = 2, 3, 4$. For $\theta < \pi/2$, let S_{θ} denote the union of the sectors in $\overline{Q_2} \cup \overline{Q_4}$ given by

$$S_{\theta} = \left\{ \xi : \frac{\pi}{2} \leq \arg \xi \leq \frac{\pi}{2} + \theta \right\} \cup \left\{ \xi : -\frac{\pi}{2} \leq \arg \xi \leq \theta - \frac{\pi}{2} \right\}. \quad (\text{A.2})$$

The analysis of $\tilde{w}$ requires integrals involving the function $\alpha_k(\xi) := -\sqrt{\xi^2 - k^2}$ for $\xi \in \mathbb{C}$, which has branch points at $\xi = \pm k$. The following definition fixes our choice of branch cuts.

Definition 9. Let $\sqrt{\cdot}$ be chosen to have the branch cut along the negative real axis, such that $\Re \sqrt{z} \geq 0$.

We define the functions α_k by

$$\alpha_k(\xi) = -\sqrt{i(\xi - k)}\sqrt{-i(\xi + k)}, \quad (\text{A.3})$$

which has a branch cut pointing vertically into the lower half plane at $-k$ and a branch cut pointing vertically into the upper half plane at k . Along the real axis where $|\xi| < k$, $\alpha_k(\xi) = i\sqrt{k^2 - \xi^2}$, and where $|\xi| > k$, $\alpha_k(\xi) = -\sqrt{\xi^2 - k^2}$. $\alpha_k(\xi)$ maps $\overline{Q}_2 \cup \overline{Q}_4$ into $\overline{Q}_2$ and in S_θ is equal to $i\sqrt{k^2 - \xi^2}$.

Returning to $\tilde{w}$, we begin by observing that for fixed y_2 and ξ , $\tilde{w}$ is the unique outgoing solution to the ordinary differential equation (ODE)

$$L_\xi[\tilde{w}] := (\partial_{x_2}^2 - \xi^2 + k^2(1 + q))\tilde{w} = -\frac{1}{2\alpha_k(\xi)}k^2q(x_2)e^{\alpha_k(\xi)|x_2 - y_2|}; \quad (\text{A.4})$$

the outgoing condition implies that $\tilde{w}'(x_2) = \text{sign}(x_2)\alpha_k(\xi)\tilde{w}(x_2)$ for all $|x_2| > d$. To analyze the behavior of solutions to this ODE, we begin by constructing the outgoing resolvent of L_ξ , denoted by $g(\xi; x_2, z_2)$, and defined so that the solution of $L_\xi[v] = f$ is given by

$$v(x_2) = \int_{-\infty}^{\infty} g(\xi; x_2, z_2)f(z_2)dz_2, \quad (\text{A.5})$$

provided f decays sufficiently rapidly at ∞ . For $x_2 > d$, L_ξ reduces to $(\partial_{x_2}^2 - \xi^2 + k^2)$ and its two linearly independent solutions are given by $e^{\pm\alpha_k(\xi)x_2}$. Following the analysis in [20, Appendix A], the outgoing resolvent can be expressed explicitly as

$$g(\xi; x_2, z_2) := \frac{1}{W(\xi)} [\tilde{u}_+(\xi; x_2)\tilde{u}_-(\xi; z_2)\chi_{z_2 \leq x_2} + \tilde{u}_-(\xi; x_2)\tilde{u}_+(\xi; z_2)\chi_{z_2 > x_2}], \quad (\text{A.6})$$

where $\tilde{u}_\pm$ are particular solutions of L_ξ satisfying

$$\begin{aligned} L_\xi \tilde{u}_\pm(\xi; x_2) &= 0, & x_2 &\in \mathbb{R} \\ \tilde{u}_\pm(\xi; x_2) &= e^{\pm\alpha_k(\xi)x_2}, & \pm x_2 &> d, \end{aligned} \quad (\text{A.7})$$

W_ξ is their Wronskian given by

$$W(\xi) := \tilde{u}_-(\xi; x_2)\partial_{x_2}\tilde{u}_+(\xi; x_2) - \tilde{u}_+(\xi; x_2)\partial_{x_2}\tilde{u}_-(\xi; x_2),$$

and χ_S is the indicator function of the set S .

In the following lemma, we show that the functions $\tilde{u}_\pm$ are uniquely defined, and describe their region of analyticity.

Lemma 16. *For every $\xi \in \mathbb{C}$, $\tilde{u}_\pm$ satisfying equation (A.7) are uniquely defined. Moreover, when viewed as a function of $\zeta = \alpha_k(\xi)$, the functions $\tilde{u}_\pm$, and $\partial_{x_2}\tilde{u}_\pm$ are entire in ζ .*

Proof. The proof is standard, see [15] for example. We sketch the proof for $\tilde{u}_-$. Let

$$v(\xi; x_2) := (e^{\alpha_k(\xi)x_2}\tilde{u}_-(\xi; x_2), \partial_{x_2}e^{\alpha_k(\xi)x_2}\tilde{u}_-(\xi; x_2))^T$$

and

$$B(\zeta; x_2) := \begin{pmatrix} 0 & 1 \\ -k^2q(x_2) & 2\zeta \end{pmatrix}.$$

Then (A.7) is equivalent to

$$\partial_{x_2}v(\xi; x_2) = B(\alpha_k(\xi); x_2)v(\xi; x_2), \quad v(\xi; -d) = (1, 0),$$

which in integral form is given by

$$v(\xi; x_2) = v(\xi; -d) + \int_{-d}^{x_2} B(\alpha_k(\xi); z)v(\xi; z)dz. \quad (\text{A.8})$$

As in [15], we iterate the above Volterra integral equation, and use the initial data, to obtain

$$v(\xi; x_2) = (1, 0)^T + \sum_{j=1}^{\infty} \mathfrak{B}_{\alpha_k(\xi)}^{(j)}[(1, 0)^T](x_2), \quad (\text{A.9})$$

with

$$\mathfrak{B}_{\alpha_k(\xi)}^{(j)}[w](x_2) := \int_{-d}^{x_2} \int_{-d}^{z_1} \cdots \int_{-d}^{z_{j-1}} B(\alpha_k(\xi); z_1) \cdots B(\alpha_k(\xi); z_j) w(z_j) dz_j \cdots dz_1. \quad (\text{A.10})$$

Noting that $\|B(\zeta, z)\| \leq \sqrt{4|\zeta|^2 + k^2\|q\|_{\infty} + 1}$ we find that

$$|\mathfrak{B}_{\alpha_k(\xi)}^{(j)}[w](x_2)| \leq \frac{[(x_2 + d) \sqrt{4|\zeta|^2 + k^2\|q\|_{\infty} + 1}]^j}{j!} \|w\|_{\infty},$$

where the bound holds for both components. It is then straightforward to show that the sum in (A.9) converges uniformly for ζ bounded with

$$|v(\xi; x_2)| \leq \exp((x_2 + d) \sqrt{4|\zeta|^2 + k^2\|q\|_{\infty} + 1}).$$

More refined bounds can be found for the region with $x_2 > d$ by restarting the Volterra integral equation at $x_2 = d$. For $\Re \zeta \leq 0, \zeta \neq 0$, at the expense of a multiplicative constant independent of ζ and x_2 , we can replace x_2 in the above estimate by $\min(x_2, d)$. Since each term in (A.9) is clearly analytic in ζ , and the convergence of the sum is uniform on compact sets, the analyticity of $v(\xi; x_2)$ follows immediately.

Finally, we observe that $\tilde{u}_+(\xi; x_2) = \tilde{u}_-^{\#}(\xi; -x_2)$, where $\tilde{u}_-^{\#}$ is the solution $\tilde{u}_-$ of (A.7) with $q(x_2)$ replaced by $q(-x_2)$. Thus we have established the analyticity of $\tilde{u}_{\pm}$ and $\partial_{x_2}\tilde{u}_{\pm}$, and the proof is complete. $\square$

Since the outgoing resolvent g has $W(\xi)$ in its denominator, the zeros of the Wronskian correspond to poles of g . In the following lemma, we show that the Wronskian has finitely many zeros on the real axis, and provide a uniform lower bound on W outside of a ball in $\bar{Q}_2 \cup \bar{Q}_4$. We denote this region by $\Xi := \{\bar{Q}_2 \cup \bar{Q}_4 : |\xi| \geq \xi_{\min}\}$, assuming that q is not identically zero, we set $\xi_{\min} := k\sqrt{1 + \|q\|_{\infty}}$.

Lemma 17. *The Wronskian $W(\xi)$ has a finite number of zeros $\{\pm\xi_j\}$, all of which are real and have absolute value in the interval $[k, \xi_{\min})$. Moreover, there exists a constant $c > 0$ such that $|W(\xi)| \geq c|\xi|$ uniformly in $\xi \in \Xi$.*

The proof of this result is classical; we include it only for completeness. In particular, the bound (A.12) follows immediately from [15, Lemma 1 (i)].

Proof. The self-adjointness of the operator $\partial_{x_2}^2 + k^2(1 + q(x_2))$ implies that all the zeros of W must lie on the real or imaginary axes. However, the imaginary axis and real interval $(-k, k)$ cannot contain zeros of W , as these zeros would correspond to nontrivial outgoing solutions u to $L_{\xi}u = 0$ (see, e.g. [48, Proposition 1.4]). For $\xi \in \mathbb{R}$ satisfying $|\xi| > k$, $W(\xi) = 0$ if and only if $\tilde{u}_+(\xi; \cdot)$ is an eigenfunction of $\partial_{x_2}^2 + k^2q(x_2)$ with eigenvalue $\xi^2 - k^2$. Since q is compactly supported and piecewise continuous, the operator $\partial_{x_2}^2 + k^2q(x_2)$ has a finite number of positive eigenvalues, all of which are less than $k^2\|q\|_{\infty}$. Indeed, by the Courant-Fischer min-max principle [43], each positive eigenvalue of $\partial_{x_2}^2 + k^2q(x_2)$ is bounded above by a distinct positive eigenvalue of $\partial_{x_2}^2 + k^2\|q\|_{\infty}\chi_{x_2 \in \text{supp } q}$ (see also [49, Corollary 9.43]). Thus we have shown that W has a finite number of zeros, all of which lie in $(-\xi_{\min}, -k] \cup [k, \xi_{\min}) \subset \mathbb{R}$.

It remains to verify the growth condition for W . Observe that for any $z < -d$, $\tilde{u}_-(\xi; x_2)$ satisfies the integral equation

$$\tilde{u}_-(\xi; x_2) = e^{-\alpha_k(\xi)x_2} - \int_z^{x_2} \frac{1}{\alpha_k(\xi)} \sinh(\alpha_k(\xi)(x_2 - s)) k^2 q(s) \tilde{u}_-(\xi; s) ds,$$

where $\chi_{s>0} \sinh(\alpha_k(\xi)s)/\alpha_k(\xi)$ is the causal Green's function. In particular, setting

$$\phi_-(\xi; x_2) := e^{\alpha_k(\xi)x_2} \tilde{u}_-(\xi, x_2),$$

it follows straightforwardly that

$$\phi_-(\xi; x_2) = 1 - \int_{-\infty}^{x_2} \frac{e^{2\alpha_k(\xi)(x_2-s)} - 1}{2\alpha_k(\xi)} k^2 q(s) \phi_-(\xi; s) ds =: 1 + \mathcal{B}[\phi_-(\xi; \cdot)](x_2). \quad (\text{A.11})$$

Since $\Re \alpha_k(\xi) \leq 0$ for all $\xi \in \overline{Q}_2 \cup \overline{Q}_4$ and $\text{supp}(q) \subseteq [-d, d]$, we have

$$|\mathcal{B}[\phi_-(\xi; \cdot)](x_2)| \leq \frac{C}{|\alpha_k(\xi)|} \|\phi(\xi; \cdot)\|_{L^\infty[-d, d]}, \quad \xi \in \overline{Q}_2 \cup \overline{Q}_4, \quad x_2 \in \mathbb{R}.$$

Hence for all $\xi \in \overline{Q}_2 \cup \overline{Q}_4$ sufficiently large, $I - \mathcal{B}$ is boundedly invertible on $L^\infty(\mathbb{R})$ with $\|(I - \mathcal{B})^{-1} - I\| \leq C/|\xi|$. It then follows from (A.11) that

$$|\phi_-(\xi; x_2) - 1| \leq \frac{C}{|\xi|}, \quad \xi \in \Xi, \quad x_2 \in \mathbb{R}. \quad (\text{A.12})$$

Now, when $x_2 > d$,

$$\tilde{u}_-(\xi; x_2) = a(\xi) e^{x_2 \alpha_k(\xi)} + b(\xi) e^{-x_2 \alpha_k(\xi)}, \quad (\text{A.13})$$

and thus

$$W(\xi) = b(\xi) \alpha_k(\xi), \quad b(\xi) = \phi_-(\xi; x_2) - a(\xi) e^{2x_2 \alpha_k(\xi)}. \quad (\text{A.14})$$

By (A.12),

$$|b(\xi) - 1| \leq \inf_{x_2 \in \mathbb{R}} \left(|\phi_-(\xi; x_2) - 1| + |a(\xi) e^{2x_2 \alpha_k(\xi)}| \right) \leq \frac{C}{|\xi|} + |a(\xi)| \inf_{x_2 \in \mathbb{R}} |e^{2x_2 \alpha_k(\xi)}| = \frac{C}{|\xi|} \quad (\text{A.15})$$

uniformly in $\xi \in \Xi \cap \{\Re \xi \neq 0\}$. Indeed, for any $\xi \in \Xi$ with $\Re \xi \neq 0$, we have $\Re \alpha_k(\xi) < 0$, meaning that the last infimum in the above equation vanishes. Recalling Lemma 16 and the expression for W in (A.14), we see that $b(\xi) = W(\xi)/\alpha_k(\xi)$ is an analytic function of ξ in the strip $\{|\Re \xi| < k/2\}$, hence the bound (A.15) extends to all $\xi \in \Xi$. Since W does not vanish in Ξ , we have thus established the existence of a constant $c > 0$ such that $|W(\xi)| \geq c|\xi|$ for all $\xi \in \Xi$. $\square$

The following corollary is a straightforward consequence of Lemmas 16 and 17.

Corollary 4. *The outgoing resolvent g when viewed as a function $\zeta = \alpha_k(\xi)$ is analytic in ζ with the exception of $\zeta_j = \alpha_k(\xi_j)$.*

In the following lemma, we turn our attention to properties of solutions to $L_\xi v = f$. In the remainder of the section, let $\rho_{a,b}$, for $a < b$, denote the following piecewise linear function

$$\rho_{a,b}(y) := \begin{cases} 0, & y < a \\ y - a, & a \leq y \leq b \\ b - a, & y > b \end{cases}. \quad (\text{A.16})$$

Lemma 18. *Let $m \geq 0$ and suppose $f : \mathbb{C} \times \mathbb{R}^2 \rightarrow \mathbb{C}$ satisfies*

$$|f(\xi; x_2, y_2)| \leq \begin{cases} C x_2^m e^{\Re \alpha_k(\xi)(x_2 - d + |y_2 - d|)}, & x_2 > d \\ C e^{\Re \alpha_k(\xi)|x_2 - y_2|}, & -d \leq x_2 \leq d \\ C |x_2|^m e^{\Re \alpha_k(\xi)(|x_2 + d| + |y_2 + d|)}, & x_2 < -d \end{cases} \quad (\text{A.17})$$

uniformly in $(\xi; x_2, y_2) \in \Xi \times \mathbb{R}^2$. Then the solution v to $L_\xi v(\xi; x_2, y_2) = f(\xi; x_2, y_2)$ satisfies

$$|v(\xi; x_2, y_2)| \leq \begin{cases} C x_2^m e^{\Re \alpha_k(\xi)(x_2-d+|y_2-d|)} \left(\frac{1}{|\xi|^2} + \frac{\rho_{-d,d}(-y_2)+x_2-d}{|\xi|} \right), & x_2 > d \\ C e^{\Re \alpha_k(\xi)|x_2-y_2|} \left(\frac{1}{|\xi|^2} + \frac{\rho_{x_2,d}(y_2)+\rho_{-x_2,d}(-y_2)}{|\xi|} \right), & -d \leq x_2 \leq d \\ C |x_2^m| e^{\Re \alpha_k(\xi)(|x_2+d|+|y_2+d|)} \left(\frac{1}{|\xi|^2} + \frac{\rho_{-d,d}(y_2)+|x_2+d|}{|\xi|} \right), & x_2 < -d \end{cases} \quad (\text{A.18})$$

uniformly in $(\xi; x_2, y_2) \in \Xi \times \mathbb{R}^2$. Moreover, if $f(\xi; x_2, y_2) = \mathfrak{f}(\alpha_k(\xi); x_2, y_2)$ for some function $\mathfrak{f}(\zeta; x_2, y_2)$ which is analytic in $\zeta \in \mathbb{C} \setminus \{\alpha_k(\xi_j)\}$, then $v(\xi; x_2, y_2) = \mathfrak{v}(\alpha_k(\xi); x_2, y_2)$ with $\mathfrak{v}(\zeta; x_2, y_2)$ analytic in $\zeta \in \mathbb{C} \setminus \{\alpha_k(\xi_j)\}$.

Proof. The analyticity of v in ζ follows from (A.5), and the analyticity of f and the outgoing resolvent g .

It remains to verify the decay estimates (A.18). Observe that (A.5) implies

$$\begin{aligned} |v(\xi; x_2, y_2)| &\leq C \frac{e^{\Re \alpha_k(\xi)x_2}}{|W(\xi)|} \left(\int_{-\infty}^{-d} |z_2|^m e^{-\Re \alpha_k(\xi)z_2} e^{\Re \alpha_k(\xi)(|z_2+d|+|y_2+d|)} dz_2 \right. \\ &\quad \left. + \int_{-d}^d |\tilde{u}_-(\xi; z_2)| e^{\Re \alpha_k(\xi)|z_2-y_2|} dz_2 + \int_d^{x_2} z_2^m |\tilde{u}_-(\xi; z_2)| e^{\Re \alpha_k(\xi)(z_2-d+|y_2-d|)} dz_2 \right) \\ &\quad + C \frac{|\tilde{u}_-(\xi; x_2)|}{|W(\xi)|} \int_{x_2}^{\infty} z_2^m e^{\Re \alpha_k(\xi)z_2} e^{\Re \alpha_k(\xi)(z_2-d+|y_2-d|)} dz_2 \end{aligned}$$

uniformly in $(\xi; x_2, y_2) \in \Xi \times (d, \infty) \times \mathbb{R}$. By (A.12), the above factors of $\tilde{u}_-(\xi; z_2)$ are well approximated by $e^{-z_2 \alpha_k(\xi)}$, up to terms that decay faster in $|\xi|$. Therefore,

$$\begin{aligned} |v(\xi; x_2, y_2)| &\leq C \frac{e^{\Re \alpha_k(\xi)x_2}}{|W(\xi)|} \left(\int_{-\infty}^{-d} |z_2|^m e^{-\Re \alpha_k(\xi)z_2} e^{\Re \alpha_k(\xi)(|z_2+d|+|y_2+d|)} dz_2 \right. \\ &\quad \left. + \int_{-d}^d e^{-\Re \alpha_k(\xi)z_2} e^{\Re \alpha_k(\xi)|z_2-y_2|} dz_2 + \int_d^{x_2} z_2^m e^{-\Re \alpha_k(\xi)z_2} e^{\Re \alpha_k(\xi)(z_2-d+|y_2-d|)} dz_2 \right) \\ &\quad + C \frac{e^{-\Re \alpha_k(\xi)x_2}}{|W(\xi)|} \int_{x_2}^{\infty} z_2^m e^{\Re \alpha_k(\xi)z_2} e^{\Re \alpha_k(\xi)(z_2-d+|y_2-d|)} dz_2 \end{aligned}$$

uniformly in $(\xi; x_2, y_2) \in \Xi \times (d, \infty) \times \mathbb{R}$. These integrals are estimated by

$$\begin{aligned} \int_{-\infty}^{-d} |z_2|^m e^{-\Re \alpha_k(\xi)z_2} e^{\Re \alpha_k(\xi)(|z_2+d|+|y_2+d|)} dz_2 &\leq \frac{C}{|\xi|} e^{\Re \alpha_k(\xi)d} e^{\Re \alpha_k(\xi)|y_2+d|}, \\ \int_{-d}^d e^{-\Re \alpha_k(\xi)z_2} e^{\Re \alpha_k(\xi)|z_2-y_2|} dz_2 &= -\frac{1}{2\Re \alpha_k(\xi)} (e^{-\Re \alpha_k(\xi)d} e^{\Re \alpha_k(\xi)|y_2-d|} - e^{\Re \alpha_k(\xi)d} e^{\Re \alpha_k(\xi)|y_2+d|}) + \rho_{-d,d}(-y_2) e^{-y_2 \Re \alpha_k(\xi)}, \quad (\text{A.19}) \\ \int_d^{x_2} z_2^m e^{-\Re \alpha_k(\xi)z_2} e^{\Re \alpha_k(\xi)(z_2-d+|y_2-d|)} dz_2 &\leq x_2^m (x_2 - d) e^{-\Re \alpha_k(\xi)d} e^{\Re \alpha_k(\xi)|y_2-d|}, \\ \int_{x_2}^{\infty} z_2^m e^{\Re \alpha_k(\xi)z_2} e^{\Re \alpha_k(\xi)(z_2-d+|y_2-d|)} dz_2 &\leq \frac{C}{|\xi|} x_2^m e^{-\Re \alpha_k(\xi)d} e^{2\Re \alpha_k(\xi)x_2} e^{\Re \alpha_k(\xi)|y_2-d|}, \end{aligned}$$

which implies that

$$|v(\xi; x_2, y_2)| \leq \frac{C}{|W(\xi)|} e^{\Re \alpha_k(\xi)(x_2-d+|y_2-d|)} \left(\frac{x_2^m}{|\xi|} + \rho_{-d,d}(-y_2) + x_2^m (x_2 - d) \right)$$

uniformly in $(\xi; x_2, y_2) \in \Xi \times (d, \infty) \times \mathbb{R}$. The bound in (A.18) for $x_2 > d$ then follows from the growth condition $|W(\xi)| \geq c|\xi|$ established in Lemma 17.

Applying this result to the function $v(\xi; -x_2, -y_2)$, we obtain the desired bound for $x_2 < -d$. It remains to establish (A.18) when $-d \leq x_2 \leq d$. To this end, we again use (A.5) to write

$$\begin{aligned} |v(\xi; x_2, y_2)| &\leq C \frac{|\tilde{u}_+(\xi; x_2)|}{|W(\xi)|} \left(\int_{-\infty}^{-d} |z_2|^m e^{-\Re \alpha_k(\xi) z_2} e^{\Re \alpha_k(\xi)(|z_2+d|+|y_2+d|)} dz_2 \right. \\ &\quad \left. + \int_{-d}^{x_2} |\tilde{u}_-(\xi; z_2)| e^{\Re \alpha_k(\xi)|z_2-y_2|} dz_2 \right) \\ &\quad + C \frac{|\tilde{u}_-(\xi; x_2)|}{|W(\xi)|} \left(\int_{x_2}^d |\tilde{u}_+(\xi; z_2)| e^{\Re \alpha_k(\xi)|z_2-y_2|} dz_2 \right. \\ &\quad \left. + \int_d^{\infty} z_2^m e^{\Re \alpha_k(\xi) z_2} e^{\Re \alpha_k(\xi)(z_2-d+|y_2-d|)} dz_2 \right), \end{aligned}$$

and apply (A.12) and the analogous bound for $\tilde{u}_+$ to obtain

$$\begin{aligned} |v(\xi; x_2, y_2)| &\leq C \frac{e^{\Re \alpha_k(\xi) x_2}}{|W(\xi)|} \left(\int_{-\infty}^{-d} |z_2|^m e^{-\Re \alpha_k(\xi) z_2} e^{\Re \alpha_k(\xi)(|z_2+d|+|y_2+d|)} dz_2 \right. \\ &\quad \left. + \int_{-d}^{x_2} e^{-\Re \alpha_k(\xi) z_2} e^{\Re \alpha_k(\xi)|z_2-y_2|} dz_2 \right) \\ &\quad + C \frac{e^{-\Re \alpha_k(\xi) x_2}}{|W(\xi)|} \left(\int_{x_2}^d e^{\Re \alpha_k(\xi) z_2} e^{\Re \alpha_k(\xi)|z_2-y_2|} dz_2 \right. \\ &\quad \left. + \int_d^{\infty} z_2^m e^{\Re \alpha_k(\xi) z_2} e^{\Re \alpha_k(\xi)(z_2-d+|y_2-d|)} dz_2 \right) \end{aligned} \quad (\text{A.20})$$

uniformly in $(\xi; x_2, y_2) \in \Xi \times [-d, d] \times \mathbb{R}$. A direct calculation reveals that

$$\begin{aligned} &\int_{-d}^{x_2} e^{-\Re \alpha_k(\xi) z_2} e^{\Re \alpha_k(\xi)|z_2-y_2|} dz_2 \\ &= -\frac{1}{2\Re \alpha_k(\xi)} e^{\Re \alpha_k(\xi) y_2} \left(e^{-2\Re \alpha_k(\xi) \min\{x_2, y_2\}} - e^{2\Re \alpha_k(\xi) d} \right) \chi_{(-d, \infty)}(y_2) + e^{-\Re \alpha_k(\xi) y_2} \rho_{-x_2, d}(-y_2), \end{aligned}$$

where χ_I is the indicator function of the set I . Since $\rho_{-x_2, d}(-y_2)$ vanishes for all $y_2 > x_2$, it follows that

$$\begin{aligned} &e^{\Re \alpha_k(\xi) x_2} \int_{-d}^{x_2} e^{-\Re \alpha_k(\xi) z_2} e^{\Re \alpha_k(\xi)|z_2-y_2|} dz_2 \\ &= -\frac{1}{2\Re \alpha_k(\xi)} \left(e^{\Re \alpha_k(\xi)|x_2-y_2|} - e^{\Re \alpha_k(\xi)(x_2+y_2+2d)} \right) \chi_{(-d, \infty)}(y_2) + e^{\Re \alpha_k(\xi)|x_2-y_2|} \rho_{-x_2, d}(-y_2). \end{aligned}$$

The first integral in (A.20) was already controlled in (A.19), while the third and fourth integrals in (A.20) evaluated at (x_2, y_2) are the same as the first and second evaluated at $(-x_2, -y_2)$. Combining these bounds and again using that $W(\xi) \geq c|\xi|$, we obtain

$$\begin{aligned} |v(\xi; x_2, y_2)| &\leq \frac{C}{|\xi|^2} \left(e^{\Re \alpha_k(\xi)|x_2-y_2|} + e^{\Re \alpha_k(\xi)(x_2+d+|y_2+d|)} + e^{\Re \alpha_k(\xi)(-x_2+d+|y_2-d|)} \right) \\ &\quad + \frac{C}{|\xi|} e^{\Re \alpha_k(\xi)|x_2-y_2|} (\rho_{x_2, d}(y_2) + \rho_{-x_2, d}(-y_2)) \end{aligned}$$

uniformly in $(\xi; x_2, y_2) \in \Xi \times [-d, d] \times \mathbb{R}$. Since $\pm x_2 + d + |y_2 \pm d| \geq |x_2 - y_2|$ for all $-d \leq x_2 \leq d$, we have established the remaining bound in (A.18) and the result is complete. $\square$

We now turn our attention to the main result of this appendix which summarizes estimates on the partial Fourier transform of the resolvent kernel.

Theorem 8. Suppose $q : \mathbb{R} \rightarrow [0, \infty)$ is piecewise continuous with $\text{supp}(q) \subset [-d, d]$. For fixed $(x_2, y_2) \in \mathbb{R}^2$, the partial Fourier transform $\tilde{w}(\xi; x_2, y_2)$ of the resolvent kernel has a finite set of real poles denoted by $\{\pm\xi_j : j = 1, 2, \dots, n_p\}$, all of which have modulus greater than k . Furthermore, it takes the form

$$\tilde{w}(\xi; x_2, y_2) = e^{\alpha_k(\xi)(|x_2 \mp d| + |y_2 \mp d|)} A_{\pm}(\xi, y_2), \quad A_{\pm}(\xi, y_2) = \begin{cases} A_{\pm,+}(\xi), & y_2 > d \\ A_{\pm,0}(\xi, y_2), & |y_2| \leq d \\ A_{\pm,-}(\xi), & y_2 < -d \end{cases} \quad (\text{A.21})$$

for all $\pm x_2 > d$, where the $A_{\pm}$ are of the form

$$A_{\pm}(\xi, y_2) = \frac{1}{\alpha_k(\xi)} B_{\pm}(\alpha_k(\xi), y_2), \quad \xi \in \mathbb{C}, \quad y_2 \in \mathbb{R}, \quad (\text{A.22})$$

with the functions $B_{\pm}(\zeta, y_2)$ analytic in $\zeta \in \mathbb{C} \setminus \{\alpha_k(\xi_j)\}$. For any $j \geq 0$, the functions $\partial_{\zeta}^j B_{\pm}(\zeta, y_2)$ are uniformly bounded over compact subsets of $(\mathbb{C} \setminus \{\alpha_k(\xi_j)\}) \times \mathbb{R}$. Moreover, there exists a constant $\xi_{\min} > 0$ such that,

$$|\partial_{\xi}^j A_{\pm}(\xi, y_2)| \leq \frac{C_j}{|\xi|^3} (1 + |\xi| \rho_{-d,d}(\mp y_2)), \quad \xi \in \Xi, \quad y_2 \in \mathbb{R}. \quad (\text{A.23})$$

Finally, when $-d \leq x_2 \leq d$ and $-d \leq y_2 \leq d$,

$$\tilde{w}(\xi; x_2, y_2) = e^{\alpha_k(\xi)|x_2 - y_2|} A_{0,0}(\xi, x_2, y_2),$$

where $A_{0,0}(\xi, x_2, y_2) = B_{0,0}(\alpha_k(\xi), x_2, y_2)/\alpha_k(\xi)$ with $B_{0,0}(\zeta, x_2, y_2)$ analytic in $\zeta \in \mathbb{C} \setminus \{\alpha_k(\xi_j)\}$ and $\partial_{\zeta}^j B_{0,0}(\zeta, x_2, y_2)$ uniformly bounded over compact subsets of $(\mathbb{C} \setminus \{\alpha_k(\xi_j)\}) \times \mathbb{R}$. For any $j \in \mathbb{N}_0$,

$$|\partial_{\xi}^j A_{0,0}(\xi, x_2, y_2)| \leq \frac{C_j}{|\xi|^3} (1 + |\xi|(\rho_{x_2,d}(y_2) + \rho_{-x_2,d}(-y_2))) \quad (\text{A.24})$$

uniformly in $\xi \in \Xi$ and $(x_2, y_2) \in [-d, d]^2$.

Proof. Recall that the function $\tilde{w}(\xi; x_2, y_2)$ satisfies $L_{\xi} \tilde{w}(\xi; x_2, y_2) = f(\xi; x_2, y_2)/\alpha_k(\xi)$, where

$$f(\xi; x_2, y_2) := -\frac{1}{2} k^2 q(x_2) e^{\alpha_k(\xi)|x_2 - y_2|}$$

satisfies the assumptions of Lemma 18 with $m = 0$. It follows that $\alpha_k(\xi) \tilde{w}(\xi; x_2, y_2)$ is analytic in $\alpha_k(\xi) \in \mathbb{C} \setminus \{\alpha_k(\xi_j)\}$.

Since q is compactly supported in $[-d, d]$, it follows from (A.5) and the definition of f that $\tilde{w}(\xi; x_2, y_2) e^{-(|x_2 \mp d| + |y_2 \mp d|)\alpha_k(\xi)}$ is independent of $\pm x_2 > d$, and constant in y_2 over the intervals $y_2 > d$ and $y_2 < -d$. We conclude that

$$\tilde{w}(\xi; x_2, y_2) = e^{\alpha_k(\xi)(|x_2 \mp d| + |y_2 \mp d|)} A_{\pm}(\xi, y_2), \quad \pm x_2 > d, \quad y_2 \in \mathbb{R}, \quad (\text{A.25})$$

where the $A_{\pm}(\xi, y_2)$ are constant in y_2 over the intervals $y_2 > d$ and $y_2 < -d$, and are of the form (A.22) with the $\partial_{\zeta}^j B_{\pm}(\zeta, y_2)$ locally bounded by continuity. Lemma 18 then implies that

$$|A_{\pm}(\xi, y_2)| \leq \frac{C}{|\xi|^3} (1 + |\xi| \rho_{-d,d}(\mp y_2)), \quad \xi \in \Xi, \quad y_2 \in \mathbb{R}.$$

The analogous properties of $A_{0,0}$ follow from parallel arguments, thus we have verified that $\alpha_k(\xi) \tilde{w}(\xi; x_2, y_2)$ satisfies the bound in (A.18).

It remains to verify that derivatives of $A_{\pm}$ and $A_{0,0}$ satisfy the same decay estimate. With

$$\tilde{w}^{(j)}(\xi; x_2, y_2) := \partial_{\xi}^j \tilde{w}(\xi; x_2, y_2),$$

we have that $L_\xi \tilde{w}^{(j)}(\xi; x_2, y_2) = f_j(\xi; x_2, y_2)/\alpha_k(\xi)$, where

$$\frac{f_j(\xi; x_2, y_2)}{\alpha_k(\xi)} := \partial_\xi^j \frac{f(\xi; x_2, y_2)}{\alpha_k(\xi)} + 2j\xi \tilde{w}^{(j-1)}(\xi; x_2, y_2) + 2\binom{j}{2} \tilde{w}^{(j-2)}(\xi; x_2, y_2).$$

Since $\partial_\xi^j(f(\xi; \cdot, y_2)/\alpha_k(\xi))$ vanishes outside $[-d, d]$ and satisfies

$$|\partial_\xi^j(f(\xi; x_2, y_2)/\alpha_k(\xi))| \leq C(1 + |y_2|)^j \frac{1}{|\xi|} e^{-\Re \alpha_k(\xi)|x_2 - y_2|}, \quad (\xi; x_2, y_2) \in \Xi \times [-d, d] \times \mathbb{R},$$

the function $(1 + |y_2|)^{-j} |\xi| \partial_\xi^j(f(\xi; x_2, y_2)/\alpha_k(\xi))$ is bounded by the right-hand side of (A.17) with $m = 0$. An inductive argument shows that $f_j/\alpha_k(\xi)$ satisfies (A.17) with $m = j - 1$, and thus $\tilde{w}^{(j)}$ satisfies (A.18) with $m = j - 1$. It then follows from (A.25) that

$$|\partial_\xi^j A_\pm(\xi, y_2)| \leq \frac{C_j}{|\xi|^3} (1 + |\xi| \rho_{-d, d}(\mp y_2)) (1 + |y_2|)^j, \quad \xi \in \Xi, \quad y_2 \in \mathbb{R}.$$

Since $A_\pm(\xi, \cdot)$ is constant in the intervals $(-\infty, -d)$ and (d, ∞) , the factor $(1 + |y_2|)^j$ can be absorbed into C_j . Similarly, the definition of $A_{0,0}$ implies (A.24) and the proof is complete. $\square$

Corollary 5. *For all $\xi \neq \alpha_k(\xi_j)$, the function $\tilde{w}(\xi; x_2, y_2)$ admits an analytic extension to $x_2, y_2 \in \Gamma_\mathbb{C} \times \Gamma_\mathbb{C}$.*

Proof. The result is an immediate consequence of equation (A.21). $\square$

B Kernel estimates

Having established properties of the partial Fourier transforms $\tilde{w}^{l,r}$, we now turn our attention to the analytical properties of $w^{l,r}$. The purpose of the section is twofold: a) to establish that the kernels k_C and k_D defined in terms of $w^{l,r}$ lie in the required $\mathcal{A}_\alpha$ spaces, and b) obtain bounds for the kernels $w^{l,r}(x_1, x_2; 0, y_2)$ that are uniform in $x_1 \in \mathbb{R}$, $|x_2| \leq L_C$, and $y_2 \in \Gamma_\mathbb{C}$. These estimates are essential for the proofs of Theorems 3 and 4.

Recall that

$$w^{l,r}(\mathbf{x}; \mathbf{y}) = \int_{\gamma_F} e^{i\xi(x_1 - y_1)} \tilde{w}^{l,r}(\xi; x_2, y_2) d\xi, \quad (\text{B.1})$$

and that

$$\begin{aligned} k_D(x_2, y_2) &= w^r(0, x_2; 0, y_2) - w^l(0, x_2; 0, y_2) = \int_{\gamma_F} (\tilde{w}^r(\xi; x_2, y_2) - \tilde{w}^l(\xi; x_2, y_2)) d\xi, \\ k_C(x_2, y_2) &= \frac{\partial^2}{\partial x_1 \partial y_1} (w^r(0, x_2; 0, y_2) - w^l(0, x_2; 0, y_2)) = \int_{\gamma_F} \xi^2 (\tilde{w}^r(\xi; x_2, y_2) - \tilde{w}^l(\xi; x_2, y_2)) d\xi. \end{aligned} \quad (\text{B.2})$$

In the following lemma, we establish the analyticity of $w^{l,r}$.

Lemma 19. *The formula (B.1) defines $w^{l,r}(x_1, x_2; y_1, y_2)$ for any $x_2, y_2 \in \Gamma_\mathbb{C}$ and $x_1, y_1 \in \mathbb{R}$. Further, $w^{l,r}(x_1, x_2; y_1, y_2)$ and their x_1 and y_1 derivatives are analytic functions of x_2 and y_2 in the interiors of $\Gamma_\mathbb{C}$.*

Proof. From Corollary 5, it is known that $\tilde{w}^{l,r}(\xi; x_2, y_2)$ can be analytically extended as a function of x_2 and y_2 from the real line to $\Gamma_\mathbb{C}$. The result then follows from a combination of dominated convergence theorem, Morera's theorem and the estimates in Theorem 8. $\square$

This lemma and (B.2) lead easily to the following corollary.

Corollary 6. *The kernels $k_C(x_2, y_2)$ and $k_D(x_2, y_2)$ extend analytically to x_2, y_2 in the interior of $\Gamma_\mathbb{C}$.*

Next we turn our attention to the analysis of the asymptotics of k_C , and k_D . The absence of the $e^{i\xi(x_1-y_1)}$ term in their definitions simplifies this analysis in comparison to the analysis of $w^{l,r}$. In order to establish that $k_D \in \mathcal{A}_{\frac{3}{2}}$, and $k_C \in \mathcal{A}_{\frac{1}{2}}$, all that remains to be shown are the asymptotic estimates 4.2 and 4.3. Both of these follow from a steepest descent argument, which we formulate as follows.

Lemma 20 (Steepest descent). *Let $A(\xi)$ have the following properties:*

- i) A is an even function of ξ .
- ii) The function $A(\xi)$ is analytic on $\mathbb{C}$, apart from the branch cuts of α_k and finitely many poles on the real axis, and $\xi = 0, \pm k$ are not poles.
- iii) There exists $p \in \mathbb{N}$ and $c > 0$ such that for each $\delta > 0$ there is a constant C_δ such that $|A(\xi)| \leq C_\delta ((1 + |\xi|)^{(p+1)} + e^{c|\Im \xi|})$ for all $\xi \in \mathbb{C} \setminus T_\delta$ where T_δ is the union of δ -balls around each of the poles and branch points.

We denote by γ_F the contour defined by $\Im \xi = -\tanh \Re \xi$, see Figure 9. For z in the first quadrant let $I(z)$ be the function defined by the contour integral

$$I(z) = \int_{\gamma_F} e^{\alpha_k(\xi)z} A(\xi) d\xi. \quad (\text{B.3})$$

Then, there exist constants $\{C_j : j = 0, 1, \dots\}$ depending on A such that, for any $n \in \mathbb{N}$, there is a constant K_n so that

$$\left| I(z) - \frac{e^{ikz}}{\sqrt{z}} \sum_{j=0}^n \frac{C_j}{z^j} \right| \leq e^{-k\Im z} \frac{K_n}{|z|^{n+\frac{3}{2}}}. \quad (\text{B.4})$$

as z tends to infinity in the first quadrant with $|z| > 4c$.

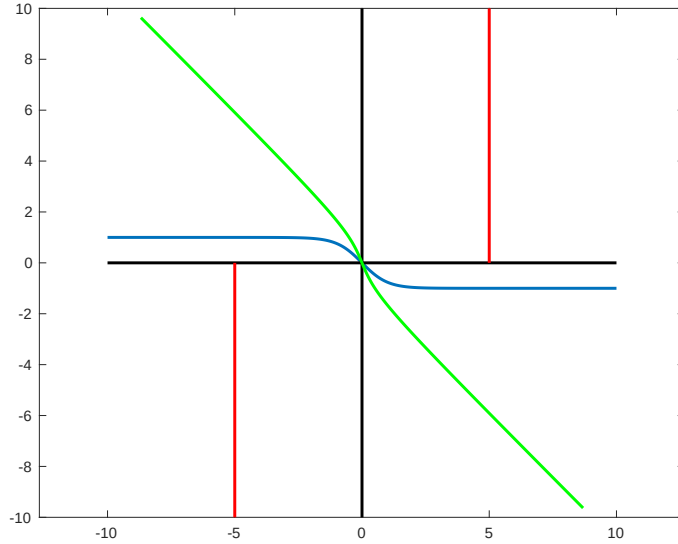


Figure 9: Plot in the ξ -plane showing γ_F in blue and $w(t)$ in green. The cuts used in the definition of $\alpha_k(\xi)$ are shown in red, for $k = 5$.

Proof. To prove this expansion for z in the first quadrant, recall that $\alpha_k(\xi)$ maps $\overline{Q}_2 \cup \overline{Q}_4$ into $\overline{Q}_2$, and therefore $\Re(\alpha_k(\xi)z) \leq 0$ for $\xi \in \overline{Q}_2 \cup \overline{Q}_4$ and $z \in \overline{Q}_1$. Suppose that w is the contour given by

$$w(t) := kt\sqrt{2}e^{-3i\pi/8}\sqrt{1 + \frac{t^2}{2}e^{i\pi/4}}, \quad (\text{B.5})$$

for $t \in \mathbb{R}$. This contour is contained in the sector $\bar{S}_{\pi/4}$, is an odd function on the real axis, is such that both $w(t)$ and $A(w(t))$ are analytic functions of t in a strip containing the real axis, and there exists a $\delta > 0$ such that $w(t)$ and $A(w(t))$ have convergent Taylor expansions centered at $t = 0$ in $|t| < \delta$. The purpose of this contour is three fold: to avoid the singularities of the integrand in $I(z)$, to ensure that the curve asymptotes at ∞ at an angle between $(\pi/2, \pi)$, and most importantly, so that the exponent $\alpha_k(w(t))$ simplifies to

$$\begin{aligned}\alpha_k(w(t)) &= i\sqrt{k^2 - w(t)^2}, \\ &= ik - kt^2 e^{-i\pi/4}.\end{aligned}\tag{B.6}$$

It is not difficult to show, using the estimates above and Cauchy's theorem, that

$$\begin{aligned}I(z) &= \int_w e^{\alpha_k z} A(\xi) d\xi \\ &= e^{ikz} \int_{-\infty}^{\infty} e^{-kt^2 z e^{-i\pi/4}} A(w(t)) w'(t) dt.\end{aligned}\tag{B.7}$$

Letting $b(t) = A(w(t))w'(t)$ and using the properties of A , and w , we note that b is an analytic function in a strip around the real axis, is an even function of t and there exist a constant C_0 such that

$$\begin{aligned}|w(t)| &\leq k(1 + t^2), \\ |b(t)| = |A(w(t))w'(t)| &\leq C_\delta \left((1 + |w|)^{(p+1)} + e^{c|\Im(w)|} \right) |w'(t)| \leq C_0 e^{2ck(1+t^2)},\end{aligned}\tag{B.8}$$

for $t \in \mathbb{R}$. Moreover, b has a Taylor expansion centered at $t = 0$ valid in $|t_0| < \delta$, i.e. there exist coefficients b_j , $j = 1, 2, \dots \infty$, and constants R_j such that

$$\left| b(t) - \sum_{j=0}^N b_j t^{2j} \right| \leq R_N |t|^{2N+2},\tag{B.9}$$

for all $|t| < \delta$. To establish the asymptotic expansions in (B.4), we split the integral into two parts $|t| < \delta$ and the remainder. For the infinite piece, we have

$$\begin{aligned}\left| \int_{\mathbb{R} \setminus [-\delta, \delta]} e^{-kt^2 z e^{-i\pi/4}} b(t) dt \right| &\leq \int_{\mathbb{R} \setminus [-\delta, \delta]} e^{-kt^2 |z|/2} |b(t)| dt, \\ &\leq C_0 e^{2ck} \int_{\mathbb{R} \setminus [-\delta, \delta]} e^{-kt^2 |z|/2 + 2ck t^2} dt, \\ &\leq C_0 e^{2ck} \frac{e^{-k\left(\frac{|z|}{2} - 2c\right)\delta^2}}{\left(\frac{|z|}{2} - 2c\right)\delta},\end{aligned}\tag{B.10}$$

where the last inequality follows from the fact that $|z| > 4c$ and [17, (7.8.3)].

For the integral on $[-\delta, \delta]$, we have

$$\int_{-\delta}^{\delta} e^{-kt^2 z e^{-i\pi/4}} b(t) dt = \int_{-\delta}^{\delta} e^{-kt^2 z e^{-i\pi/4}} \sum_{j=0}^N b_j t^{2j} dt + \int_{-\delta}^{\delta} e^{-kt^2 z e^{-i\pi/4}} r_N(t) dt,\tag{B.11}$$

where $|r_N(t)| \leq R_N |t|^{2N+2}$. Using the identity

$$\int_{-\infty}^{\infty} e^{-kt^2 z e^{-i\pi/4}} t^{2j} dt = \frac{\Gamma(j + 1/2)}{(k z e^{-i\pi/4})^{j+1/2}},\tag{B.12}$$

and an estimate similar to equation (B.10), we get

$$\int_{-\delta}^{\delta} e^{-kt^2 z e^{-i\pi/4}} \sum_{j=0}^N b_j t^{2j} dt = \sum_{j=0}^N b_j \frac{\Gamma(j + 1/2)}{(k z e^{-i\pi/4})^{j+1/2}} + O\left(e^{-\frac{|z|}{2} k \delta^2}\right).\tag{B.13}$$

The remainder term $r_n(t)$ can be bounded in a similar manner so that

$$\left| \int_{-\delta}^{\delta} e^{-kt^2 z e^{-i\pi/4}} r_N(t) dt \right| \leq \frac{C_N}{|z|^{N+3/2}}, \quad (\text{B.14})$$

for some $C_N > 0$. The proof then follows by combining equations (B.10), (B.13), and (B.14). $\square$

In the following lemma, we prove asymptotic estimates for the kernels k_C , and k_D .

Lemma 21. *Suppose $q^{l,r}$ satisfy the hypothesis in Theorem 8. Then the corresponding kernels k_C and k_D satisfy the estimates (4.2) and (4.3) with $\alpha = 3/2$ and $1/2$ respectively.*

Proof. As an example, we prove the estimate for k_D when $x_2, y_2 \in \Gamma_R$. The other cases follow by nearly identical proofs. From Theorem 8, it follows that there exists functions $A_{++},r(\xi)$ and $A_{++},l(\xi)$ such that $\tilde{w}^{l,r}(\xi; x_2, y_2) = A_{++},l,r(\xi) e^{\alpha_k(\xi)(x_2+y_2-2d)}$. This implies that

$$k_D(x_2, y_2) = \int_{\gamma_F} (A_{++},r(\xi) - A_{++},l(\xi)) e^{\alpha_k(\xi)(x_2+y_2-2d)} d\xi. \quad (\text{B.15})$$

Furthermore, Theorem 8 also implies that $A_{++},r(\xi) - A_{++},l(\xi)$ satisfy the conditions in Lemma 20. The result then follows from Lemma 20 since $x_2 + y_2 - 2d \in \bar{Q}_1$.

Similarly, the kernel k_C has the form

$$k_C(x_2, y_2) = e^{ik(x_2+y_2-2d)} \int_{\gamma_F} \xi^2 (A_{++},r(\xi) - A_{++},l(\xi)) e^{-(\alpha_k + ik)(x_2+y_2-2d)} d\xi, \quad (\text{B.16})$$

which is of the form required by Lemma 20 with

$$A(\xi) = \xi^2 (A_{++},r(\xi) - A_{++},l(\xi)). \quad (\text{B.17})$$

Since this has a second order zero at $\xi = 0$, the $j = 0$ term in (B.4) vanishes. Repeating this argument for the other regions gives the desired result. $\square$

Next we turn our attention to the asymptotic behavior of $w^{l,r}$. In the remainder, for notational convenience, we will assume that $y_1 = 0$, since the kernels are only a function of $x_1 - y_1$. Referring to equation (B.1), we note that for any finite x_1, y_1 , the functions $w^{l,r}$ agree with the construction in [20]. However, the contour γ_F is not suitable for analyzing the asymptotics as $|x_1| \rightarrow \infty$ as it involves values of ξ with $\Im \xi$ of both signs. For these calculations one needs to deform to the contours $\Gamma_{\nu}^{\pm}$ for $\pm x_1 > 0$, introduced in [20], see Figure 10. These deformations require crossing poles of $w^{l,r}(\xi; x_2, y_2)$, which leads to a residue contribution composed of waveguide modes: for $\pm x_1 > 0$ we have with $l \leftrightarrow -, r \leftrightarrow +$,

$$w^{l,r}(x_1, x_2; 0, y_2) = \int_{\Gamma_{\nu}^{\pm}} e^{i\xi x_1} \tilde{w}^{l,r}(\xi; x_2, y_2) d\xi + \sum_{j=1}^{N_{l,r}} v_j^{\pm}(x_2) v_j^{\pm}(y_2) e^{\pm i\xi_j^{\pm} x_1}. \quad (\text{B.18})$$

In this representation the contributions to the integral from non-real ξ decay exponentially as $\pm x_1 \rightarrow \infty$. Where $\pm x_2 \gg d$ the functions $v_j^{\pm}(x_2) = c_{j,\pm}^{\pm} e^{\mp x_2 \sqrt{(\xi_j^{\pm})^2 - k^2}}$, and so obviously have analytic continuations to $\Gamma_{\mathbb{C}}$, which decay exponentially as $|\Re x_2| \rightarrow \infty$, but do not decay as $|\Im x_2| \rightarrow \infty$. This leads to the restriction that the data belongs to a space $\mathcal{C}^{\alpha,\beta} \oplus \mathcal{C}^{\alpha+\frac{1}{2},\beta}$, with $\beta \geq 0$, and the restriction that y_2 lie on a contour, $\tilde{\Gamma}$, that satisfies the slope condition in (3.12). In the following lemma, we establish the asymptotic properties of $w^{l,r}$.

Lemma 22. *Suppose that $y_2 \in \tilde{\Gamma} \subset \mathcal{G}$ where $\tilde{\Gamma}$ satisfies the slope condition in (3.12). Then for any $n \in \mathbb{N}_0$,*

$$|w^{l,r}(x_1, x_2; 0, y_2)| + |\partial_{x_1} w^{l,r}(x_1, x_2; 0, y_2)| \leq C_n \frac{1 + |x_1|^n + |y_2|^n}{1 + |y_2|^{2n+1}} \quad (\text{B.19})$$

uniformly in $\mathbf{x} \in \mathbb{R}^2 \cap \{|x_2| \leq L_C\}$ and $y_2 \in \tilde{\Gamma}$.

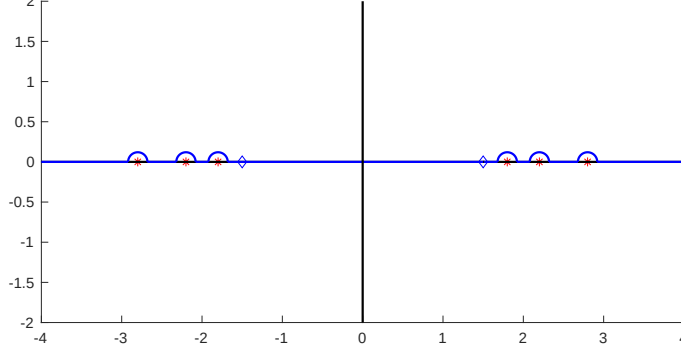


Figure 10: Plot in the ξ -plane showing Γ_ν^+ in blue. The zeros of the Wronskian are marked by red stars, and the branch points are marked by diamonds.

Proof. Note that various decay estimates for $w^{l,r}(x_1, x_2; 0, y_2)$ with x_1, x_2 and y_2 real were already established in [21]. All that remains to be shown is that the estimates continue to hold when $y_2 \in \tilde{\Gamma}$. Assume $x_1 > 0$ for concreteness. We recall (B.18), which states that

$$w(x_1, x_2; 0, y_2) = w_0(x_1, x_2; y_2) + w_{\text{wg}}(x_1, x_2; y_2)$$

where

$$w_0(x_1, x_2; y_2) = \int_{\Gamma_\nu^+} e^{i\xi x_1} \tilde{w}(\xi; x_2, y_2) d\xi, \quad w_{\text{wg}}(x_1, x_2; y_2) = \sum_{j=1}^{N_{\text{wg}}} v_j(x_2) v_j(y_2) e^{i\xi_j x_1} \quad (\text{B.20})$$

and we have dropped the superscripts l, r for convenience. The waveguide contribution w_{wg} decays exponentially in $|\Re y_2|$ and thus clearly satisfies (B.19).

We next consider w_0 . By the decay of $\tilde{w}(\cdot; x_2, y_2)$ established in Theorem 8,

$$\int_{\Gamma_\nu^+} |\tilde{w}(\xi; x_2, y_2)| d\xi + \int_{\Gamma_\nu^+} |\xi \tilde{w}(\xi; x_2, y_2)| d\xi$$

is a continuous function of (x_2, y_2) . Since $\Im \xi \geq 0$ for all $\xi \in \Gamma_\nu^+$, this means that $w_0(x_1, x_2; y_2)$ and $\partial_{x_1} w_0(x_1, x_2; y_2)$ are bounded uniformly in $x_1 \in \mathbb{R}$ over (x_2, y_2) in compact sets. It remains to bound $w_0(x_1, x_2; y_2)$ and $\partial_{x_1} w_0(x_1, x_2; y_2)$ for $y_2 \in \tilde{\Gamma} \cap \{|\Re y_2| > d\} \cap \{c|\Re y_2| \leq |\Im y_2| \leq C|\Re y_2|\}$.

Using the shorthand $z := |x_2 - d| + \sqrt{(y_2 - d)^2}$, we invoke Theorem 8 to write

$$\tilde{w}(\xi; x_2, y_2) = e^{-\alpha_k(\xi)z} A(\xi, x_2), \quad \xi \in \Gamma_\nu^+, \quad x_2 \in \mathbb{R}, \quad y_2 \in \tilde{\Gamma} \cap \{\Re y_2 > d\},$$

where $A(\xi, x_2) = B(\alpha_k(\xi), x_2)/\alpha_k(\xi)$ with $B(\zeta, x_2)$ analytic in $\zeta \in \mathbb{C} \setminus \{\alpha_k(\xi_j)\}$. We estimate the integral in (B.20) by separately handling the region where $\alpha_k(\xi)$ is small. To this end, fix $0 < \delta < \min\{k, \xi_1 - \nu - k\}$ and let $\psi \in \mathcal{C}_c^\infty((-k - \delta, -k + \delta) \cup (k - \delta, k + \delta))$ such that $\psi \equiv 1$ in $(-k - \delta/2, -k + \delta/2) \cup (k - \delta/2, k + \delta/2)$. This definition ensures that Γ_ν^+ is real on the support of ψ , and we extend ψ to be 0 on the parts of Γ_ν^+ that are not real. We write $w_0 = w_{0,0} + w_{0,1}$, where

$$w_{0,1}(x_1, x_2; y_2) := \int_{\Gamma_\nu^+} e^{i\xi x_1} \tilde{w}(\xi; x_2, y_2) \psi(\xi) d\xi.$$

Looking first at $w_{0,0}$, since $\Re \alpha_k(\xi) \geq \sqrt{\delta k - \delta^2/4} =: \eta$ and $0 \leq \Im \alpha_k(\xi) \leq 2\nu(\xi_p + \nu)$ for all $\xi \in \Gamma_\nu^+ \cap \text{supp}(1 - \psi)$,

$$|w_{0,0}(x_1, x_2; y_2)| \leq \int_{\Gamma_\nu^+} e^{-\Re \alpha_k(\xi) \Re y_2 + \Im \alpha_k(\xi) \Im y_2} |A(\xi, x_2)(1 - \psi(\xi))| d\xi \leq C e^{2\nu(\xi_p + \nu) \Im y_2 - \eta \Re y_2} \quad (\text{B.21})$$

uniformly in $\mathbf{x} \in \mathbb{R}^2 \cap \{x_1 > 0\} \cap \{|x_2| \leq N\}$ and $y_2 \in \tilde{\Gamma} \cap \{\Re y_2 > d\}$.

The remaining contribution is given by

$$w_{0,1}(x_1, x_2; y_2) = \int_{\Gamma_v^+} e^{i\xi x_1 - \alpha_k(\xi)y_2} \frac{B(\alpha_k(\xi), x_2)}{\alpha_k(\xi)} \psi(\xi) d\xi.$$

After a change of variables in the integral, this becomes

$$\begin{aligned} w_{0,1}(x_1, x_2; y_2) &= 2 \int_0^{\eta_+} \cos(\sqrt{\zeta^2 + k^2} x_1) e^{-\zeta y_2} \frac{B(\zeta, x_2)}{\sqrt{\zeta^2 + k^2}} \psi(\sqrt{\zeta^2 + k^2}) d\zeta \\ &\quad + 2i \int_0^{\eta_-} \cos(\sqrt{k^2 - \zeta^2} x_1) e^{i\zeta y_2} \frac{B(-i\zeta, x_2)}{\sqrt{k^2 - \zeta^2}} \psi(\sqrt{k^2 - \zeta^2}) d\zeta, \end{aligned}$$

where we have assumed without loss of generality that ψ is even, and have defined $\eta_+ := \sqrt{(k + \delta)^2 - k^2}$ and $\eta_- := \sqrt{k^2 - (k - \delta)^2}$. It is clear that the first term on the right-hand side is bounded by $C/\Re y_2$, with the second bounded by $C/\Im y_2$. We note that almost identical arguments apply to the case for which $y_2 \in \tilde{\Gamma} \cap \{\Re y_2 < -d\}$ and $x_1 < 0$. Enforcing the comparability constraint $c|\Re y_2| \leq |\Im y_2| \leq C|\Re y_2|$, and choosing $\nu > 0$ sufficiently small, (B.21) together with the above bounds imply that

$$|w_0(x_1, x_2; y_2)| \leq \frac{C}{1 + |y_2|}, \quad \mathbf{x} \in \mathbb{R}^2 \cap \{|x_2| \leq L_C\}, \quad y_2 \in \tilde{\Gamma}.$$

It is clear that the above arguments extend to $\partial_{x_1} w_0$, thus we have verified (B.19) when $n = 0$.

To extend this result to arbitrary n , we define $\varphi(\xi, \zeta, x_1, y_2) := i\xi x_1 - \zeta y_2$ and use that

$$\partial_\xi \varphi(\xi, \alpha_k(\xi), x_1, y_2) = ix_1 - \alpha'_k(\xi) y_2 = \varphi(\alpha_k(\xi), \xi, x_1, y_2) / \alpha_k(\xi)$$

to write

$$w_{0,1}(x_1, x_2; y_2) = \int_{\Gamma_v^+} \frac{B(\alpha_k(\xi), x_2) \psi(\xi)}{\alpha_k(\xi)} \left(\frac{\alpha_k(\xi)}{\varphi(\alpha_k(\xi), \xi, x_1, y_2)} \partial_\xi \right)^n e^{\varphi(\xi, \alpha_k(\xi), x_1, y_2)} d\xi.$$

Now, observe that since $\alpha_k(\mathbb{R}) \subset \mathbb{R} \cup i\mathbb{R}$, the function $\varphi(\alpha_k(\xi), \xi, x_1, y_2)$ is nonzero for all $(\xi, x_1, y_2) \in \text{supp}(\psi) \times \mathbb{R} \times Q_1$, and $\partial_\xi \varphi(\alpha_k(\xi), \xi, x_1, y_2) = \varphi(\xi, \alpha_k(\xi), x_1, y_2) / \alpha_k(\xi)$ is locally integrable. We stress here that the comparability constraint implies that $\arg y_2$ is bounded away from zero in the above integral. Therefore, we can integrate by parts to obtain

$$w_{0,1}(x_1, x_2; y_2) = (-1)^n \int_{\Gamma_v^+} e^{\varphi(\xi, \alpha_k(\xi), x_1, y_2)} \left(\partial_\xi \frac{\alpha_k(\xi)}{\varphi(\alpha_k(\xi), \xi, x_1, y_2)} \right)^n \frac{B(\alpha_k(\xi), x_2) \psi(\xi)}{\alpha_k(\xi)} d\xi,$$

which by the local boundedness of the $\partial_\xi^n B(\zeta, x_2)$ can be written as

$$w_{0,1}(x_1, x_2; y_2) = \int_{\Gamma_v^+} e^{\varphi(\xi, \alpha_k(\xi), x_1, y_2)} \frac{1}{\alpha_k(\xi)} \sum_{j=0}^n \frac{\phi_j(\xi, x_1, y_2)}{\varphi^{n+j}(\alpha_k(\xi), \xi, x_1, y_2)} d\xi,$$

where the ϕ_j are compactly supported in ξ and satisfy $|\phi_j(\xi, x_1, y_2)| \leq C(1 + |x_1|^j + |y_2|^j)$ uniformly in all variables. As above, we change variables in the integral to obtain

$$\begin{aligned} w_{0,1}(x_1, x_2; y_2) &= \sum_{\varepsilon \in \{-1, 1\}} \left\{ \int_0^{\eta_+} e^{\varepsilon i \sqrt{\zeta^2 + k^2} x_1} e^{-\zeta y_2} \frac{1}{\sqrt{\zeta^2 + k^2}} \sum_{j=0}^n \frac{\phi_j(\varepsilon \sqrt{\zeta^2 + k^2}, x_1, y_2)}{\varphi^{n+j}(\zeta, \varepsilon \sqrt{\zeta^2 + k^2}, x_1, y_2)} d\zeta \right. \\ &\quad \left. + i \int_0^{\eta_-} e^{\varepsilon i \sqrt{k^2 - \zeta^2} x_1} e^{i\zeta y_2} \frac{1}{\sqrt{k^2 - \zeta^2}} \sum_{j=0}^n \frac{\phi_j(\varepsilon \sqrt{k^2 - \zeta^2}, x_1, y_2)}{\varphi^{n+j}(-i\zeta, \varepsilon \sqrt{k^2 - \zeta^2}, x_1, y_2)} d\zeta \right\}. \end{aligned}$$

Since $|1/\varphi(\zeta, \varepsilon\sqrt{\zeta^2 + k^2}, x_1, y_2)| + |1/\varphi(-i\zeta, \varepsilon\sqrt{k^2 - \zeta^2}, x_1, y_2)| \leq C/|y_2|$ uniformly in $0 < \zeta < \eta_+$,

$$|w_{0,1}(x_1, x_2; y_2)| \leq C \sum_{j=0}^n \frac{1 + |x_1|^j + |y_2|^j}{|y_2|^{n+j}} \left(\int_0^{\eta_+} e^{-\zeta \Re y_2} d\zeta + \int_0^{\eta_-} e^{-\zeta \Im y_2} d\zeta \right) \leq C \frac{1 + |x_1|^n + |y_2|^n}{|y_2|^{2n+1}},$$

where the last inequality follows from the comparability assumption $c|\Re y_2| \leq |\Im y_2| \leq C|\Re y_2|$. Given the exponential decay of $w_{0,0}$ in (B.21) with $\nu > 0$ sufficiently small, we have now established that

$$|w_0(x_1, x_2; y_2)| \leq C_n \frac{1 + |x_1|^n + |y_2|^n}{1 + |y_2|^{2n+1}}, \quad \mathbf{x} \in \mathbb{R}^2 \cap \{|x_2| \leq L_C\}, \quad y_2 \in \tilde{\Gamma}.$$

It is again clear that the same estimates apply to $\partial_{x_1} w_0$, and the proof is complete. $\square$

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